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Biometric sensor system and method for monitoring a driver of a vehicle

Abstract

A driver monitoring (DM) computing device for monitoring driving behavior of a driver in real-time is provided. The DM computing device detects a user computer device associated with a driver inside the vehicle. The DM computing device collects state data of the user computer device. The state data includes data as to a state of the user computer device during a currently occurring trip. The DM computing device collects vehicle operation data of the vehicle while the driver and the user computer device are inside the vehicle. The DM computing device compares the collected vehicle operation data and the state data of the user computer device to determine whether the user computer device was activated during the currently occurring trip. The DM computing device causes a driving reward to be applied to an account associated with the driver for inactivity of the user computer device during the currently occurring trip.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation-in-part of and claims priority to U.S. patent application Ser. No. 16/122,635, filed Sep. 5, 2018, and entitled “DRIVER MONITORING SYSTEM AND METHOD,” the contents and disclosures of which are hereby incorporated by reference in their entirety.

BACKGROUND

(1) This disclosure relates generally to driver monitoring systems and, more specifically, to network-based systems and methods for monitoring a driver's driving behavior in real-time.

(2) Many vehicle accidents occur due to drivers being distracted by their phones while driving. Although drivers may know the risks associated with phone use while driving (e.g., causing a

vehicle accident), many drivers continue to talk on the phone, send text messages, and/or browse applications on their phone while driving. Vehicle accidents can be costly, time consuming, and in serious cases, fatal. Although many states ban certain types of phone use while driving (e.g., texting and/or using a handheld phone), phone use in its entirety is not yet prohibited. Thus, drivers, especially young drivers, are more prone to browse through their phones (e.g., to check messages, go on social media, and/or take and upload photos) while they are on the road. Although some vehicle monitoring systems are known, they are susceptible to providing delayed and/or inaccurate data in regards to a driver's driving behavior. Such known vehicle monitoring systems do not reward and/or penalize a bank account and/or a payment card account associated with the driver based on the driver's monitored driving behavior. Thus, there exists a need for a real-time driver monitoring system that can provide accurate data as to a driver's driving behavior, and provide rewards and/or penalties to incentivize safe driving practices.

(3) Additionally, new drivers generally must undergo training and testing before being granted driving credentials, such as a driver's license. While vehicle monitoring systems may generate data relevant to determining whether a driver has qualified for certain driving credentials, such known vehicle monitoring systems generally are not capable of making such determinations. Further, once driving credentials are granted to a new driver, there is no way to control access to these credentials, for example, based on an ongoing analysis of the driver's driving. Thus, there exists a need for a real-time driver monitoring system that can provide accurate data as to a driver's driving behavior, and control access to driving credentials based on this data.

BRIEF DESCRIPTION OF THE DISCLOSURE

(4) In one aspect, a DM computing device for monitoring driving behavior is provided. The DM computing device may include a processor in communication with a memory and a vehicle. The processor may be configured to store, in a database, baseline biometric information for a plurality of registered drivers. The processor may be further configured to collect biometric data generated by one or more biometric sensors of the vehicle. The processor may be further configured to compare the collected biometric data to the baseline biometric information to identify a first driver of the plurality of registered drivers that is present in a vehicle. The processor may be further configured to collect vehicle operation data of the vehicle while the first driver is in the vehicle. The processor may be further configured to store the collected vehicle operation data in association with the first driver in the database. The processor may be further configured to generate, by applying an artificial intelligence model to the collected vehicle operation data stored in association with the first driver, at least one driver assessment value. The processor may be further configured to determine the first driver qualifies for a driving credential by comparing the at least one driver assessment value to at least one threshold. The processor may be further configured ton in response to the determination, provide access to a digital document within a virtual wallet application executed by a user device associated with the first driver, the digital document associated with the driving credential.

(5) In another aspect, a computer-implemented method for monitoring driving behavior is provided. The computer-implemented method may be performed by a DM computing device including a processor in communication with a memory and a vehicle. The computer-implemented method may include storing, in a database, baseline biometric information for a plurality of registered drivers. The computer-implemented method may further include collecting biometric data generated by one or more biometric sensors of the vehicle. The computer-implemented method may further include comparing the collected biometric data to the baseline biometric information to identify a first driver of the plurality of registered drivers that is present in a vehicle. The computer-implemented method may further include collecting vehicle operation data of the vehicle while the first driver is in the vehicle. The computer-implemented method may further include storing the collected vehicle operation data in association with the first driver in the database. The computer-implemented method may further include generating, by applying an

artificial intelligence model to the collected vehicle operation data stored in association with the first driver, at least one driver assessment value. The computer-implemented method may further include determining the first driver qualifies for a driving credential by comparing the at least one driver assessment value to at least one threshold. The computer-implemented method may further include in response to the determination, providing access to a digital document within a virtual wallet application executed by a user device associated with the first driver, the digital document associated with the driving credential.

(6) In yet another aspect, at least one non-transitory computer-readable media having computer-executable instructions embodied thereon is provided. When executed by a DM computing device including at least one processor in communication with a memory and a vehicle, the computer-executable instructions may cause the at least one processor to store, in a database, baseline biometric information for a plurality of registered drivers. The computer-executable instructions may further cause the at least one processor to collect biometric data generated by one or more biometric sensors of the vehicle. The computer-executable instructions may further cause the at least one processor to compare the collected biometric data to the baseline biometric information to identify a first driver of the plurality of registered drivers that is present in a vehicle. The computer-executable instructions may further cause the at least one processor to collect vehicle operation data of the vehicle while the first driver is in the vehicle. The computer-executable instructions may further cause the at least one processor to store the collected vehicle operation data in association with the first driver in the database. The computer-executable instructions may further cause the at least one processor to generate, by applying an artificial intelligence model to the collected vehicle operation data stored in association with the first driver, at least one driver assessment value. The computer-executable instructions may further cause the at least one processor to determine the first driver qualifies for a driving credential by comparing the at least one driver assessment value to at least one threshold. The computer-executable instructions may further cause the at least one processor to, in response to the determination, provide access to a digital document within a virtual wallet application executed by a user device associated with the first driver, the digital document associated with the driving credential.

(7) In yet another aspect, a driver monitoring (DM) computing device includes a processor and a memory in communication with the processor. The processor is programmed to detect a user computer device associated with a driver inside the vehicle. The processor is further programmed to collect state data of the user computer device. The state data includes data as to a state of the user computer device during a currently occurring trip. The state is at least one of an inactive state and an active state. The processor is further programmed to collect vehicle operation data of the vehicle while the driver and the user computer device are inside the vehicle, the vehicle operation data including vehicle telematics data. The processor is also programmed to compare the collected vehicle operation data and the collected state data of the user computer device to determine whether the user computer device was activated during the currently occurring trip. The processor is further programmed to cause a driving reward to be applied to an account associated with the driver for inactivity of the user computer device during the currently occurring trip, the inactivity being based on the inactive state.

(8) In yet another aspect, a computer-implemented method for monitoring driving behavior of a driver in real-time may be provided. The method may be implemented using a driver monitoring (DM) computing device. The DM computing device may include a processor in communication with a memory device. The method includes detecting a user computer device associated with the driver inside the vehicle, wherein the user computer device is in an inactive state. The method further includes collecting state data of the user computer device, the state data including data as to a state of the user computer device during a currently occurring trip. The state is at least one of an inactive state and an active state. The method also includes collecting vehicle operation data of the vehicle while the driver and the user computer device are inside the vehicle. The vehicle operation

data includes vehicle telematics data. The method further includes comparing the collected vehicle operation data and the collected state data of the user computer device to determine whether the user computer device was activated during the currently occurring trip. The method also includes causing a driving reward to be applied to an account associated with the driver for inactivity of the user computer device during the currently occurring trip, the inactivity being based on the inactive state.

(9) In yet another aspect, a non-transitory computer-readable storage media having computer-executable instructions embodied thereon is provided. When executed by a driver monitoring (DM) computing device including a processor coupled to a memory, the computer-executable instructions cause the DM computing device to detect a user computer device associated with a driver inside the vehicle. The computer-executable instructions further cause the DM computing device to collect state data of the user computer device. The state data includes data as to a state of the user computer device during a currently occurring trip. The state being at least one of an inactive state and an active state. The computer-executable instructions further cause the DM computing device to collect vehicle operation data of the vehicle while the driver and the user computer device are inside the vehicle. The vehicle operation data includes vehicle telematics data. The computer-executable instructions further cause the DM computing device to compare the collected vehicle operation data and the collected state data of the user computer device to determine whether the user computer device was activated during the currently occurring trip. The computer-executable instructions further cause the DM computing device to cause a driving reward to be applied to an account associated with the driver for inactivity of the user computer device during the currently occurring trip, the inactivity being based on the inactive state.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIGS. **1**-**8**B show example embodiments of the methods and systems described herein.

(2) FIG. **1** illustrates a schematic diagram of an example vehicle having a driver monitoring (DM) computing device in accordance with the present disclosure.

(3) FIG. **2** is a simplified block diagram of an example driver monitoring (DM) computer system as shown in FIG. **1** for implementing the processes shown in FIGS. **4**, **8**A, and **8**B, in accordance with the present disclosure.

(4) FIG. **3** illustrates an example use case for monitoring a driver's driving behavior using the DM computing device shown in FIG. **1**, in accordance with the present disclosure.

(5) FIG. **4** illustrates a flow chart of an example process for monitoring driving behavior of a driver using the DM computing device shown in FIG. **1**, in accordance with the present disclosure.

(6) FIG. **5** illustrates an example configuration of a user computer device for use in the system shown in FIG. **2**.

(7) FIG. **6** illustrates an example configuration of a server computer device for use in the system shown in FIG. **2**.

(8) FIG. **7** illustrates a diagram of components of one or more example computing devices that may be used in the computing system shown in FIG. **2**.

(9) FIG. **8**A illustrates a flow chart of another example process for monitoring driving behavior of a driver using the DM computing device shown in FIG. **1**, in accordance with the present disclosure.

(10) FIG. **8**B is a continuation of the flow chart illustrated in FIG. **8**A.

(11) Like numbers in the Figures indicate the same or functionally similar components. Although specific features of various embodiments may be shown in some figures and not in others, this is for convenience only. Any feature of any figure may be referenced and/or claimed in combination with any feature of any other figure.

DETAILED DESCRIPTION OF THE DISCLOSURE

(12) Systems and methods according to this disclosure are directed to monitoring real-time driving behavior, and more specifically, to detect usage of a user computer device (e.g., mobile device) by a driver during a driving event or trip (e.g., whenever a driver is driving a vehicle). In the example embodiment, a driver monitoring (DM) computing device (e.g., a DM server) is in communication with a vehicle controller and a user computer device (e.g., a mobile device) associated with a driver. As used herein, the term “vehicle controller” refers generally to a vehicle's electronic control unit (ECU) that controls one or more electrical systems and/or subsystems of a vehicle. The vehicle controller is in communication with the vehicle's sensors to monitor all aspects of the vehicle's engine operation (e.g., measuring and adjusting electronic valve control, speed, engine temperature, ignition timing, vehicle emissions control). The vehicle controller collects on-board diagnostics data as to the status of the vehicle's engine and various vehicle systems. The vehicle controller may also collect vehicle operation data including telematics data from the vehicle's sensors and/or various systems, such as an in-car entertainment (ICE) system or an in-vehicle infotainment (IVI) system that manages audio content, navigation, traffic conditions, weather, entertainment (e.g., movies games, radio), phone calls, and/or text messages. In the example embodiment, the DM computing device is a backend remote server (e.g., DM server) that is configured to receive data associated with the vehicle and the driver from the vehicle controller and/or the user computer device. A DM module on the user computer device and/or the vehicle controller enables the vehicle controller and/or the user computer device to communicate with the DM computing device. The DM module is remote from the DM computing device. In the example embodiment, the DM module is a software application on the user computer device. In some embodiments, the remote DM module is a plug-in to the vehicle controller. In other embodiments, the remote DM module is a separate device that is embedded in the vehicle. In other words, in the example embodiment, telematics data can be captured by the vehicle controller and/or the user computer device, and transmitted to the DM computing device via the remote DM module.

(13) In the example embodiment, the vehicle may possess autonomous or semi-autonomous technology or functionality to enable biometric identification. The DM computing device receives biometric data from a user via a biometric input device associated with the vehicle. As used herein, the term “biometric data” refers generally to the field of biometrics or biometric identification/authentication, in which characteristics or traits of humans are captured and analyzed, often for distinguishing one individual from another using a uniqueness of the characteristics or traits captured. Some examples of biometric data include fingerprints, DNA, facial images, retinal images, and voiceprints. Biometric data may be collected by sensors, such as biometric sensors (e.g., biometric input devices) on the vehicle. Biometric sensors may include any sensor configured to receive a biological signal uniquely identifying an individual, such as, but not limited to, retinal scanners, fingerprint scanners, facial recognition devices, voice input devices, and weight scales. In the example embodiment, the vehicle may have one or more biometric sensors inside the vehicle to facilitate the sampling of biometric data from users of the vehicle. Biometric input devices may include, but are not limited, to fingerprint scanners on a vehicle component near the driver, such as the dashboard, the console, or the steering wheel. In some embodiments, the biometric input device is a biometric steering wheel that continuously feeds biometric data of a user (e.g., a driver) to the DM computing device throughout an active trip (e.g., a currently occurring trip). In another embodiment, the biometric input device is a camera that monitors a driver of the vehicle. In other embodiments, the biometric input device only receives biometric data from the user prior to the initiation of a trip.

(14) In the example embodiment, the DM computing device subsequently compares received biometric data (e.g., sample data) from the user within the vehicle to a baseline biometric data stored in the memory. As used herein, “baseline biometric data” refers to reference data of authorized drivers of the vehicle to which sample data received from users of the vehicle can be

compared. Authorized drivers of the vehicle may include those listed as a named insured and/or an authorized driver under an automobile insurance policy for vehicle. In the example embodiment, authorized drivers have previously provided biometric data, such as fingerprints, facial scans, and/or retinal scans to the DM computing device, and the provided biometric data is used as baseline biometric data. In some embodiments, authorized users may have previously provided baseline biometric data to an insurance provider of the automobile insurance policy associated with the vehicle. In these embodiments, the DM computing device may be in communication with the insurance provider via insurance network to access the baseline biometric data.

(15) In the example embodiment, the DM computing device determines, from the comparison, that the current vehicle user is an authorized driver. If the received sample data matches the stored baseline biometric data, the DM computing device recognizes the vehicle user as an authorized driver and associates the active trip with the driver. If the DM computing device cannot recognize the vehicle user as an authorized driver, the DM computing device may prompt the vehicle user to provide a different type of sample data and/or to try again. If the vehicle user is not an authorized driver, the DM computing device may not associate the active trip with the vehicle user. In the example embodiment, the DM computing device detects a user computer device, such as a mobile device associated with the driver inside the vehicle. The DM computing device may communicate with the user computer device through the DM module stored on the vehicle or on the user computer device, and, for example, via the Internet, Bluetooth®, or any other wired or wireless connection (e.g., Near Field Communication) over one or more radio links or wireless communication channels. For example, the user computer device may automatically pair with a vehicle system (e.g., vehicle controller) via Bluetooth® and the DM computing device may communicate with the vehicle controller to identify the user computer device. In some embodiments, upon detecting the user computer device, the DM computing device causes the user computer device to automatically activate (e.g., launch, open) an application (e.g., the DM module) on the user computer device. The application detects a state (e.g., inactive and active) of the user computer device. In some embodiments, the application automatically switches the user computer device from an active state to an inactive state before the trip.

(16) In the example embodiment, the DM computing device communicates with the user computer device to monitor a state of the user computer device during an active trip. The DM computing device monitors and/or detects whether the user computer device is in an active state or an inactive state by using GPS, cellular triangulation, and biometrics data of the driver. As used herein, “inactive state” refers to a locked mode where the screen of the user computer device is locked. In some embodiments, applications may be running in the background, such as location services and/or the activated application, while the user computer device is in an inactive state. In locked mode (e.g., inactive state), the user computer device may be awake but displaying minimum information (e.g., time and date on a locked screen or home screen). When awake, the user computer device displays the locked screen or home screen, and receives incoming communications. To switch from the inactive state to an active state, the driver needs to actively swipe the screen, provide a physical unlock gesture (e.g., moving and/or shaking the mobile device), and/or provide identification information (e.g., passcode, unlock pattern, and/or biometric data) to access the applications and contents of the user computer device. In some embodiments, the inactive state may also include a child mode and/or a hands-free mode. As used herein, “child mode” refers to a setting offered by the user computer device that restricts usage to kid friendly, parent-approved applications. Thus, the DM computing device may recognize that the driver's user computer device is inactive by detecting the setting (e.g., do not disturb mode, child mode, driving mode) of the user computer device. In one example, the DM computing device may determine that someone is playing a game on the driver's user computer device. In this example, the DM computing device may determine that the user computer device is in an inactive state for the driver's trip because the DM computing device concurrently detects both of the driver's hands on a

biometric steering wheel.

(17) “Usage data” as referred to herein includes information as to the inactive state and the active state during a trip. For example, usage data includes a time and/or a distance the driver's user computer device spends in each state during the active trip. In another example, usage data provides information as to how often the user computer device switches between the two states during an active trip. For example, for a 30 minute trip, the DM computing device may determine that the user computer device was in an inactive state for the first 20 minutes of the trip, and in an active state for the remaining 10 minutes of the trip. In this example, the DM computing device may further determine that during the inactive state, the driver received an incoming call and held a conversation for 20 minutes. The DM computing device may categorize the incoming call and the subsequent conversation as the mobile device being in an inactive state by detecting that the call was received through the vehicle's Bluetooth® system rather than through the driver's mobile device. The DM computing device may further determine the geographical location where the incoming call was received (e.g., map coordinates, school zone, construction zone, parking lot) by utilizing a GPS-related system such as a vehicle location system and/or a mobile device locator. In the example embodiment, the driving data and the user device data may be communicated to the DM computing device through at least one of the vehicle controller, the user computer device, and the DM module.

(18) In some embodiments, driving data associated with the driver's driving characteristics may also be tracked in addition to monitoring the state of the driver's user computer device during a trip. In some embodiments where the vehicle is a semi-autonomous or a regular vehicle (e.g., driver-controlled vehicle), such that a driver controls the vehicle during part or the entirety of one or more trips, the DM computing device interprets sensor information to collect and/or generate telematics data (e.g., driving data) associated with driving characteristics of the driver. In some embodiments, the DM computing device includes the DM module that is stored on or part of a mobile device, such as the driver's phone. In these embodiments, the DM module may obtain telematics data from the vehicle via Bluetooth®, Internet, Universal Serial Bus (USB), On-Board Diagnostics (OBDII) port, or any other wired or wireless connection. For example, the vehicle controller may collect vehicle telematics data from the driver's user computer device and/or one or more sensors on the vehicle. Vehicle telematics data may include data from the user computer device and/or one or more of vehicle sensors, and may include navigation, communications, safety, security, and/or “infotainment” data. For example, vehicle telematics data collected and analyzed may include, but is not limited to, braking and/or acceleration data, navigation data, vehicle settings, and/or any other telematics data. This vehicle telematics data can then be provided to the DM computing device through at least one of the vehicle controller, the user computer device, and the DM module.

(19) In further embodiments, the driver is rewarded for having the user computer device in an inactive state during a trip. The DM computing device may calculate driving rewards based on the length at which the driver's user computer device is in an inactive state for a trip. In some embodiments, the DM computing device determines, from the received telematics data, that the driver has stopped the vehicle to use the user computer device (e.g., mobile device). For example, the DM computing device determines from the vehicle sensors, such as GPS and/or a vehicle's tracking system, that the driver has pulled the vehicle off to the side of the road and/or made a brief stop at, for example, a gas station, rest stop, restaurant, and/or convenience store during an active trip to a final destination. In this example, the DM computing device determines that (i) the vehicle is parked (e.g., in a parking mode), and (ii) the mobile device has switched from the inactive state to the active state. Based on the detection, in these embodiments, the DM computing device rewards the driver for stopping the vehicle to use the mobile device rather than using the mobile device while the vehicle is in motion. In other embodiments, the DM computing device rewards the driver for waiting to use the mobile device until the vehicle is in a parking mode, even though the driver has not yet reached the final destination (e.g., finished the active trip). The DM computing

device uses telematics data to distinguish between parking mode (e.g., where a vehicle is parked) and temporary halt motions of the vehicle while the driver waits at an intersection, red light, and/or stop sign (e.g., where the driver's foot is actively pressing on the brake pedal). Driving rewards are calculated based on safe driving practices in accordance with standards and guidelines, such as those provided by the National Transportation Safety Board (NTSB) and the National Highway Traffic Safety Administration (NHTSA).

(20) Driving rewards can be based on at least one of distance traveled and duration. Driving rewards may be calculated periodically (e.g., daily, weekly, monthly) based upon a pre-determined timeframe. Driving rewards may be a credit or discount applicable to an insurance payment account associated with an insurance policy of the driver (e.g., monthly discounts applicable to a driver's next statement). Additionally or alternatively, driving rewards may be redeemable for cash. In these embodiments, the DM computing device may transfer the driving rewards via direct deposit to a payment account (e.g., credit account, debit account, and/or prepaid account) associated with the driver. In certain embodiments, the driving rewards may be redeemable as a prepaid credit and/or debit card such as a reloadable prepaid MASTERCARD® gift card. Reward payouts may depend on factors such as the driver's payout preferences (e.g., weekly, monthly, quarterly and/or semi-annual payouts), and the driver's reward redemption preference (e.g., an insurance credit and/or discount applicable to a driver's automobile insurance statement, a direct deposit into a payment account associated with the driver, a reloadable prepaid gift card). For example, a dependent may drive a vehicle owned by a parent. As used herein, the term “dependent” refers to a child, teenager, and/or any other dependent associated with the parent who owns the vehicle. In this example, the parent may use the system described herein to reward the dependent when the dependent drives the vehicle without texting, talking on their smartphone, or otherwise using their smartphone while driving. The reward may take the form of a payment to an account or prepaid payment card associated with the dependent. In addition, the parent could also use the system described herein to penalize the dependent if the dependent texts, talks on their smartphone, or otherwise uses their smartphone while driving. The penalty may take the form of removing funds from an account associated with the dependent and/or freezing a payment card (e.g., temporarily placing a hold on a payment card) associated with the dependent for a period of time.

(21) In further embodiments, the driver is penalized for having the user computer device in an active state during the trip. In these embodiments, the driver may be penalized for switching the user computer device from an inactive state to an active state while the vehicle is in motion (e.g., during the active trip). Additionally or alternatively, the driver may be penalized for the duration over which the user computer device is detected to be in an active state. For example, the DM computing device may determine that the driver's mobile device was switched from an inactive state to an active state 3 times during the trip. In this example, the DM computing device penalizes the driver by reducing the driver's pre-payout accumulated driving rewards (for safe driving behavior) by a predefined amount and/or percentage. For example, if the driver elected to receive a credit on their next insurance policy statement, the credit amount can be reduced due to the driver's use of their mobile device during active trips. In other embodiments, the DM computing device penalizes the driver by disabling certain functions of the driver's mobile device for a designated period of time. In these embodiments, the DM computing device transmits a signal to an application on the driver's mobile device to disable and/or partially disable the mobile device.

(22) In further embodiments, the DM computing device penalizes the driver by deactivating a payment account associated with the driver for a designated period of time. In these embodiments, the DM computing device has a payment account associated with the driver stored in its database. For example, the payment account information may previously have been provided by the driver during a registration process, and/or by an insurer provider associated with the driver to the DM computing device. In another embodiment, the DM computing device may penalize the driver by receiving automatic payments (e.g., payment transfers) from a payment account associated with the

driver each time the driver's mobile device is detected to be in an active state. In these embodiments, the driver selects a penalty option where, upon detecting use of the driver's mobile device during an active trip, the DM computing device transmits a payment request for a predetermined amount to a payment network associated with a payment account tied to the driver. The DM computing device may transmit a payment request each time the driver's mobile device is detected to be in an active state. For example, during a registration process, the driver authorizes automatic payments of $5 for every time the driver's mobile device is detected to be in an active state during an active trip. In certain embodiments, the DM computing device transmits a payment request for a total number of active state detections during a predetermined period of time. In the above example, if the DM computing device detected the driver's mobile device in an active state 10 times over a week, the DM computing device transmits a single payment request at the end of the week for $50 to a payment network associated with the driver's payment account.
(23) In further embodiments, the payment account linked with the DM computing device may be that of a named insured, rather than an authorized driver listed on an automobile insurance policy associated with the vehicle. For example, if the named insured is a parent, the linked payment account may be that of the parent's. In this example, the parent's payment account is subject to a driving reward and/or driving penalty (e.g., driving disincentive). Thus, a dependent, such as a teenage driver, listed as an authorized driver under the parent's automobile insurance policy may impact the parent's payment account. For example, if the dependent repeatedly uses their mobile device over a period of time, the DM computing device may temporarily deactivate a payment account associated with the parent. In some embodiments, the DM computing device penalizes the driver based on factors such as the duration of the detected active state and/or the number of occurrences in which the driver's mobile device switches from an inactive state to an active state. For example, if the detected active state is a first time offense, the DM computing device may generate and transmit a warning (e.g., alert message) to the driver. For example, the generated warning may alert the driver of the detected active state, and warn the driver that a penalty will be enforced if the active state is detected a second time.
(24) The DM computing device may further generate a report that includes information as to the state(s) of the user computer device during a trip, the driving data described above, and/or the calculated rewards. The report may be generated daily, weekly, and/or monthly, and the reporting period may depend on how often the driver uses the vehicle. The DM computing device transmits the generated report to the driver's user computer device and/or to the driver via another communication channel, and in some embodiments, the DM computing device also transmits the generated report to a remote-computing device associated with an insurance provider. For example, the generated report may be transmitted to an insurance provider associated with the driver to allow the insurance provider to review the calculations and distribute the driving reward. In another example, where the driver is a new and/or teenage driver insured under a parent's insurance policy, the generated report may be transmitted to the parent of the driver. In another example, the generated report may be transmitted to an insurance provider to generate and/or update a driving profile associated with the driver. The information in the generated report may be used by the insurance provider to adjust the driver's insurance premium.
(25) In an exemplary embodiment, the DM computing device may be used by agency or organization responsible for issuing driving credentials (e.g., driver's licenses or other driving qualifications or endorsements) to determine whether an individual driver qualifies for these driving credentials based on vehicle operation data relating to the individual's driving. The DM computing device may analyze vehicle operation data corresponding to a driver to determine one or more values relating to a safety or quality of the driver's driving. For example, the values may include and/or relate to a total amount of time driven, speed (e.g., with respect to a speed limit), acceleration, braking, turning, use or status of the user computing device while driving, use of turn signals, proximity to other vehicles or objects, other driving events, and/or scores derived from

and/or calculated based on such data. The DM computing device may compare these values to thresholds to determine if the driver qualifies for a driving credential (e.g., a drivers license), and if so, may automatically provide access to a digital document relating to these credentials (e.g., a digital driver's license) within a virtual wallet application associated with the driver (e.g., at a user device of the driver). After the virtual document has been issued, the DM computing device may continually monitor subsequent vehicle operation data associated with the driver and control access to the virtual document based on whether driving assessment values associated with the driver continue to meet the corresponding thresholds. The DM computing device may also use the artificial intelligence model to generate reports and/or recommendations for areas to work on to assist the driver in qualifying for the driving credentials.

(26) In the exemplary embodiment, a driver desiring to obtain a driving credential may register via the DM computing device (e.g., upon obtaining a practice or learner's permit), which may be operated by or on behalf of a Department of Motor Vehicles (DMV) or other entity responsible for issuing driving credentials. During the registration, the DM computing device may be configured to store, in a database, identities and/or identifiers associated with one or more registered drivers and baseline biometric information for each of the registered drivers. As described above, this baseline biometric information enables the DM computing device to determine when a registered driver is operating a vehicle. In addition to biometric data, other data relating to the registered driver may also be stored. This may include demographic data (e.g., age, location) or other data based on which a determination may be made to grant the registered driver any driving credentials. For example, in some embodiments, thresholds used to determine whether a registered driver qualifies may depend on an age of the registered driver (e.g., a sixteen-year-old may need to meet a higher set of standards than an eighteen-year-old).

(27) In the exemplary embodiment, when a driver begins to operate a vehicle, the DM computing device may receive biometric data from the vehicle and compare biometric data received from the vehicle to the baseline biometric information to identify a current driver from the plurality of registered drivers. As described above, the vehicle may be configured to collect various types of biometric data, such as image data captured for face identification and/or fingerprint data captured for fingerprint identification. In some embodiments, the DM computing device may continuously collect biometric data while the vehicle is being operated to verify that the same driver is continuing to operate the vehicle and/or determine if a new driver has taken over operation of the vehicle. When the current driver is identified, the DM computing device may log the trip and any vehicle operation data received during the trip in the database in association with the current driver. In some embodiments, in addition to or rather than biometric data, other types of data may be used to identify the current driver, such as a manual entry of data and/or scanning a bar code or other visual code (e.g., on a practice or learner's permit or displayed by the user device).

(28) In the exemplary embodiment, the DM computing device may be configured to collect vehicle operation data of the vehicle while the identified driver is in the vehicle (e.g., during the trip). The vehicle operation data collected may include various different metrics, such as those described above, which may be used to assess the driver's performance and determine whether the driver has met any qualifications of obtaining a driving credential.

(29) In some embodiments, the collected vehicle operation data may include speed data indicative of a speed of the vehicle at a certain time and location. This speed data may be generated by a speedometer, determined based on tracking a geolocation of the vehicle (e.g., using GPS), and/or determined in other ways. The speed data may also include a speed limit at the corresponding time and location. For example, the DM computing device may perform a lookup to determine a speed limit at a give location at which a speed measurement was made. Thus, at any given time, it may be determined whether the vehicle is traveling above or below the speed limit and by how much. This speed data may be used as a factor in determining whether the driver qualifies for a driving credential.

(30) In some embodiments, the collected vehicle operation data may include proximity data obtained from exterior radar sensors, lidar sensors, cameras, and/or other devices capable of determining how close the vehicle is to other vehicles and objects. This data may indicate whether the driver is driving safely. For example, the proximity data may indicate whether the driver is maintaining a safe distance between their vehicle and a preceding vehicle (e.g., depending on a speed the vehicles are traveling, or may indicate whether the vehicle is traveling dangerously close to stationary objects (e.g., roadside objects or debris on the road) at unsafe speeds. Additionally, proximity data may be used to assess parking, for example, by determining an adequate amount of space is maintained around the vehicle during and after parking. This proximity data may be used as a factor in determining whether the driver qualifies for a driving credential.
(31) In some embodiments, the collected vehicle operation data may include interior camera data obtained from an interior camera of the vehicle. This interior camera data may indicate whether the driver is engaging in safe or unsafe practices while driving. For example, the interior camera data may indicate whether the driver is maintaining attention to the road, periodically checking mirrors and/or checking mirrors before lane changes, and/or whether the driver is becoming distracted while driving. This interior camera data may be used as a factor in determining whether the driver qualifies for a driving credential.
(32) In some embodiments, the collected vehicle operation data may include turn signal data. For example, the vehicle may be configured to report when its turn signals are being operated. This data may be compared with camera data and/or location data to determine whether the driver is activating the turn signals before turning or changing lanes. This turn signal data may be used as a factor in determining whether the driver qualifies for a driving credential.
(33) In some embodiments, the collected vehicle operation data may include other types of data. For example, any of the other types of data described herein, such as a total amount of time or distance the driver has undergone instruction, state data of a user computing device of the driver and/or other telematics data may be used to determine whether the driver is safely operating the vehicle and/or the driver qualifies for a driving credential.
(34) In the example embodiment, the DM computing device may be configured to generate, by applying an artificial intelligence model to the collected vehicle operation data stored in association with the first driver, at least one driver assessment value. The driver assessment value may correspond to a parameter such as total amount of time or distance driven, or may be another type of score or value derived from the vehicle operation data that indicates, for example, how safe the driver has driven and/or how well the driver has mastered certain aspects of driving. For example, scores may be provided for a total amount of time driven, speed (e.g., with respect to a speed limit), acceleration, braking, turning, use or status of the user computing device while driving, use of turn signals, proximity to other vehicles or objects, other driving events, and/or scores derived from a combination of these factors, such as an overall driving or safety score. The generated driver assessment values may be stored in the database in association with the corresponding driver.
(35) The DM computing device may be configured to train the artificial intelligence model using sample data (e.g., pairs of sets of sample vehicle operation data and sets of sample driver assessment values that should result from the sample vehicle operation data), historical data, and/or some combination thereof. In some embodiments, the artificial intelligence model may be periodically or continuously retrained as new data (e.g., vehicle operation data and associated outcomes) becomes available. Alternatively, other supervised or unsupervised machine learning techniques may be used to train such a model.
(36) In the exemplary embodiment, the DM computing device may be configured to determine a registered driver qualifies for a driving credential by comparing the at least one driver assessment value associated with the driver to at least one threshold corresponding to the driving credential. For example, the thresholds may specify a total time or distance that needs to be driven during driving instruction and/or certain minimum scores relating to driving safety and/or driving abilities

(e.g., parking and/or driving in high-traffic areas). In some embodiments, the specific thresholds used to determine whether a particular driver qualify for the driving credential may depend on other information relating to the driver. For example, in some cases, the thresholds may depend on the age of the driver, in which a younger driver (e.g., a sixteen-year-old) may be subject to more stringent thresholds than an older driver (e.g., an eighteen-year-old). In another example, the thresholds may depend on a location where the driver lives and/or has practiced driving. For example, the thresholds may depend on a jurisdiction in which the driver lives, whether the driver lives in an urban, suburban, and/or rural area, and/or whether the driver lives in an area deemed to be safer or less safe for driving. In some embodiments, the thresholds may be based on a comparison to other drivers. For example, a driver may qualify if their driver assessment values are better than a certain predefined percentage of other drivers. In addition to driving credentials, these driver assessment values and thresholds may be used to determine if the driver qualifies for other driving rewards as described herein.

(37) In the exemplary embodiment, the DM computing device may be configured to, in response to a determination that a driver has qualified for a driving credential, provide access to a digital document associated with the driving credential. For example, the driver may have a user device configured to execute a virtual wallet application. If the driver qualifies for a driving credential, the DM computing device may communicate with the user device to cause a digital document (e.g., a virtual driver license) to automatically be provided within the virtual wallet, thus enabling the driver to obtain instant proof of their driving credential. In some embodiment, the digital document being present in the virtual wallet may enable the driver to unlock or activate (e.g., using an NFC, Bluetooth, or other wireless connection) vehicles to which the driver has authorized access.

(38) In some embodiments, the DM computing device may control access to the digital document, such as by locking or removing the digital document from the virtual wallet, based on an ongoing analysis of the driver's driving. For example, the DM computing device may continue to collect vehicle operation data after the driver has earned the driving credential to determine updated driver assessment values. If the updated driver assessment values fail to meet the required thresholds, the DM computing device may determine to, for example, temporarily suspend or revoke the driving credentials and/or impose certain restrictions (e.g., locations or times of day where the driver can drive unaccompanied) until the thresholds are met. In such cases, the DM computing device may automatically remove the digital document from the virtual wallet and/or modify the digital document to indicate that it has a locked or restricted status.

(39) In some embodiments, the DM computing device may be further configured to determine a cost, such as a cost associated with obtaining the driving credential, based on the at least one driver assessment value. For example, different fee levels may be associated with obtaining the driving credential and/or the corresponding digital document, each with different associated thresholds, and the DM computing device may determine the fee based on which of these thresholds are met by the corresponding driver assessment values. Alternatively, the cost may be calculated as a function of the one or more driver assessment values. Other costs associated with the vehicle such as, for example, registration fees, personal property tax, insurance costs, emergency service costs, and/or other costs may be similarly determined or calculated based on the driver assessment values associated with the driver. In cases where these costs are recurring, updated driver assessment values based on new vehicle operation data may be used each time the cost is determined.

(40) In some embodiments, the DM computing device may be further configured to generate, by applying the artificial intelligence model to the collected vehicle operation data stored in association with the first driver, at least one recommendation for improving driving skills. For example, the DM computing device may determine that, for a driver, a driver assessment value and/or score generated by the artificial intelligence model that is associated with parallel parking needs to be improved before the driver can qualify for driving credentials. The DM computing device may then generate a recommendation that the driver practice parallel parking. The DM

computing device may similarly identify other areas of strength or areas of improvements, such as reducing speeding, leaving greater distance between vehicles, reducing distractions or usage of the user device during driving, checking mirrors or blind spots, or any other steps to driver safer and potentially qualify for driving credentials. The driver may then focus on these areas when practicing driving. Such recommendations may be displayed within a report provided by the DM to the user, for example, within a web page and/or app accessible through the driver's user device. The report may also include driver assessment values, vehicle operation data, charts, graphs, links to instructional videos, and/or other information that may assist the driver in reaching a driving proficiency necessary for obtaining a desired driver credential. In certain embodiments, the reports may also include data that enables a driver to compare their scores to others, such as average for other practicing drivers in the area or for drivers as a whole. Such recommendations and reports may also be generated after the driver has obtained the desired driver credential, so that the driver may continue with driving skill improvement and maintaining any earned driver credentials. In some embodiments, these recommendations and reports may be generated and updated in real time based on newly available data.

(41) In some embodiments, the DM computing device may analyze groups of drivers, such as all the drivers trained by a particular driving instructor in order to assess and/or generate recommendations for the driving instructor. Such an analysis may be performed similarly to the analysis for an individual driver, with the artificial intelligence model instead using vehicle operation data associated with a group of drivers, such as those associated with the particular driving instructor. Based on this data, the DM computing device may generate driver instructor assessment values or scores, reports, and recommendations for improvement for the driving instructor. For example, if an instructor's students tend to struggle with parallel parking, the DM computing device may generate a recommendation that the driving instructor spend more time with each student practicing parallel parking. In some embodiments, this assessment o a driving instructor may be used, for example, to determine appropriate compensation for driving instructor services and/or made available to the public to assist in selection of a driving instructor.

(42) The technical effect of the systems and methods described herein is achieved by performing at least one of the following steps: (i) detecting a user computer device associated with the driver inside the vehicle, wherein the user computer device is in an inactive state; (ii) collecting state data of the user computer device, the state data including data as to a state of the user computer device during a currently occurring trip, the state being at least one of an inactive state and an active state; (iii) collecting vehicle operation data of the vehicle while the driver and the user computer device are inside the vehicle, the vehicle operation data including vehicle telematics data; (iv) comparing the collected vehicle operation data and the collected state data of the user computer device to determine whether the user computer device was activated during the currently occurring trip; (v) causing a driving reward to be applied to an account associated with the driver for inactivity of the user computer device during the currently occurring trip, the inactivity being based on the inactive state; (vi) having the DM computing device communicate with the vehicle (e.g., via the vehicle's vehicle controller), the user computer device, and a payment network to transfer funds (e.g., driving rewards) to a payment account associated with the driver; and (vi) storing driver data associated with each driver of the vehicle, the driver data including vehicle operation data, payment account data, and driving reward data

(43) The resulting technical effect achieved by the systems and methods described herein is at least one of: (i) accurately monitoring the actions of a driver in real-time to ensure the driver is not driving while using a handheld mobile device such as a cell phone; (ii) accurately identifying a driver of a trip using biometric data; (iii) increasing driver compliance with safe driving practices by calculating driving rewards; (iv) improving real-time data collection of a driver's driving behavior by capturing continuous data as to a driver's usage of a handheld device during a trip; and/or (v) continuously improving the accuracy of data used to make insurance decisions; (vi)

enabling an objective computer-based determination of whether a driver should be granted driving credentials by analyzing vehicle operation data derived from vehicle sensors using an artificial intelligence model; (vii) enabling an ability to provide in real time proof of driving credentials by providing a digital document associated with the driving credentials in a virtual wallet application in response to a determination that the driving credentials should be granted; (viii) enabling an ability to controlling access to a digital document associated with driving credentials of a driver by continuously analyzing vehicle operation data of the driver using an artificial intelligence model; (ix) enabling an ability to generate recommendations for improving driving skills by analyzing vehicle operation data derived from vehicle sensors using an artificial intelligence model; and/or (x) enabling an ability to determining a cost associated with obtaining a driving credential by analyzing vehicle operation data derived from vehicle sensors using an artificial intelligence model.

(44) In one embodiment, a computer program is provided, and the program is embodied on a computer-readable medium. In an example embodiment, the system is executed on a single computer system, without requiring a connection to a server computer. In a further example embodiment, the system is being run in a Windows® environment (Windows is a registered trademark of Microsoft Corporation, Redmond, Washington). In yet another embodiment, the system is run on a mainframe environment and a UNIX® server environment (UNIX is a registered trademark of X/Open Company Limited located in Reading, Berkshire, United Kingdom). In a further embodiment, the system is run on an iOS® environment (iOS is a registered trademark of Cisco Systems, Inc. located in San Jose, CA). In yet a further embodiment, the system is run on a Mac OS® environment (Mac OS is a registered trademark of Apple Inc. located in Cupertino, CA). In still yet a further embodiment, the system is run on Android® OS (Android is a registered trademark of Google, Inc. of Mountain View, CA). In another embodiment, the system is run on Linux® OS (Linux is a registered trademark of Linus Torvalds of Boston, MA). The application is flexible and designed to run in various different environments without compromising any major functionality. The following detailed description illustrates embodiments of the disclosure by way of example and not by way of limitation. It is contemplated that the disclosure has general application to providing an on-demand ecosystem in industrial, commercial, and residential applications.

(45) As used herein, an element or step recited in the singular and preceded with the word “a” or “an” should be understood as not excluding plural elements or steps, unless such exclusion is explicitly recited. Furthermore, references to “example embodiment” or “one embodiment” of the present disclosure are not intended to be interpreted as excluding the existence of additional embodiments that also incorporate the recited features.

(46) FIG. **1** is a schematic diagram of an example vehicle **100**. In some embodiments, vehicle **100** may be an autonomous or semi-autonomous vehicle capable of fulfilling the transportation capabilities of a traditional automobile or other vehicle. In these embodiments, vehicle **100** may be capable of sensing its environment and navigating without human input. In other embodiments, vehicle **100** may be a “driver-needed” vehicle, such as a traditional automobile that is controlled by a human driver **115**.

(47) As shown in FIG. **1**, vehicle **100** may include a vehicle controller **110** associated with vehicle **100**, and a mobile device **125** associated with a vehicle user in communication with a remote driver monitoring (DM) server (e.g., DM computing device) **210**. In the example embodiment, vehicle controller **110** and/or mobile device **125** can include a DM module **130** that enables vehicle controller **110** and mobile device **125** to communicate with DM computing device **210**. Telematics data and mobile device **125** usage data can be captured by vehicle controller **110** and/or mobile device **125**, and transmitted to DM computing device **210** during a trip with the help of DM module **130**. In the example embodiment, vehicle **100** may also include and employ one or more sensors **105**, such as biometric sensors **105** to determine which vehicle user is driver **115** for a trip. Biometric sensors **105** may include any sensor configured to receive a biological signal uniquely

identifying an individual, such as, but not limited to, retinal scanners, fingerprint scanners, facial recognition devices, and weight scales. In the example embodiment, vehicle **100** may have one or more biometric input devices (e.g., biometric sensors **105**) to facilitate the sampling of biometric data from users of vehicle **100**. Biometric input devices may include, but are not limited, to fingerprint scanners on a component of vehicle **100** only easily accessible by the driver, such as the dashboard, the console, or the steering wheel. Biometric data may include fingerprints, DNA, facial images, retinal images, and/or voiceprints. In some embodiments, the biometric input device collects biometric data (e.g., sample data) and transmits the data to other system components for analysis. Additionally or alternatively, vehicle **100** may have a weight scale (or pressure sensor) associated with the driver's seat and/or with the passenger's seats. Vehicle **100** may have a registered weight associated with each user of vehicle **100**. When any vehicle user sits in any of the seats, their weight may be measured by the scale and the particular vehicle user may be identified.
(48) Vehicle **100** may also include sensors **105** that detect the current surroundings and location of vehicle **100**. Sensors **105** may include, but are not limited to, radar, LIDAR, Global Positioning System (GPS), video devices, imaging devices, cameras, audio recorders, and computer vision. Sensors **105** may also detect conditions of vehicle **100**, such as speed, acceleration, gear, braking, cornering, and other conditions related to the operation of vehicle **100**, for example: at least one of a measurement of at least one of speed, direction rate of acceleration, rate of deceleration, location, position, orientation, and rotation of the vehicle, and a measurement of one or more changes to at least one of speed, direction rate of acceleration, rate of deceleration, location, position, orientation, and rotation of the vehicle. Furthermore, sensors **105** may include impact sensors that detect impacts to vehicle **100**, including force and direction and sensors that detect actions of vehicle **100**, such the deployment of airbags.
(49) In some embodiments, in addition to detecting the presence of driver **115**, sensors **105** may also detect the presence of one or more passengers **120** in vehicle **100**. In these embodiments, sensors **105** may detect the presence of fastened seatbelts, the weight in each seat in vehicle **100**, heat signatures, or any other method of detecting information about driver **115** and passengers **120** in vehicle **100**.
(50) In certain embodiments, in addition to biometric sensors **105**, vehicle **100** may include occupant position sensors to determine a location and/or position of each occupant (e.g., driver **115** and, in some embodiments, passengers **120**) in vehicle **100**. In these embodiments, the location of an occupant may identify a particular seat or other location within vehicle **100** where the occupant is located. The position of the occupant may include the occupant's body orientation, the location of specific limbs, and/or other positional information. In one example, sensors **105** may include an in-cabin facing camera, LIDAR, radar, weight sensors, accelerometer, gyroscope, compass and/or other types of sensors to identify the location and/or position of occupants within vehicle **100**.
(51) Vehicle controller **110** may be configured to recognize driver **115** as an authorized (e.g., registered) driver from biometric data received from sensors **105** (e.g., biometric input device). Vehicle controller **110** may be configured to communicate with driver's **115** mobile device **125**, such as a smartphone. Mobile device **125** may be similar to user computer device **206** (shown in FIG. **2**). In the example embodiment, vehicle controller **110** wirelessly communicates with mobile device **125** in vehicle **100**. In these embodiments, vehicle controller **110** may be capable of communicating with mobile device **125** of a vehicle occupant, such as driver **115**, through DM module **130**. For example, DM module **130** can be a software application on mobile device **125**. In the example embodiment, upon detecting the vehicle user as driver **115**, vehicle **100**, via vehicle controller **110**, may communicate with mobile device **125**, for example, via the Internet, Bluetooth®, USB, OBDII port, or any other wired or wireless connection (e.g., Near Field Communication) over one or more radio links or wireless communication channels to automatically activate (e.g., launch) and/or communicate with DM module **130**. For example, DM module **130** may be an application that detects a state of mobile device **125** and/or switches mobile device **125**

from an active state to an inactive state. In some embodiments, vehicle **100** may have “application pairing” functionality such that driver **115** may engage with an application on a user interface at vehicle **100** while their mobile device **125** is in an inactive state. In other embodiments, vehicle **100** may be in communication with one or more mobile devices **125** that are each associated with one or more vehicle occupants of vehicle **100**.

(52) Using the application pairing functionality, vehicle controller **110** may further determine which mobile device(s) **125** are within vehicle **100** during a trip. For example, each vehicle occupant may pair one or more mobile device **125** (e.g., smartphones, tablets, laptops, wearables, etc.) to vehicle **100**. Vehicle **100** may then pair with one or more mobile device **125** that are within vehicle **100** during a trip. Vehicle controller **110** may record which device(s) **125** pair with vehicle **100** for a trip. Using the received biometric data from the biometric sensors, vehicle controller **110** may confirm that the vehicle occupant associated with the paired mobile device **125** is driver **115** for the trip.

(53) In some embodiments, vehicle controller **110** may include a communication interface device (shown in FIG. **2**) that includes a display screen or touchscreen. The communication interface device may be capable of displaying information to and receiving information from driver **115**. In some embodiments, after mobile device **125** of driver **115** pairs with vehicle **100**, select applications and functions of mobile device **125**, such as navigation, call dialer, and music may be displayed on the communication interface device of vehicle **100**. In these embodiments, mobile device **125** may be in a locked (e.g., inactive) state, and driver **115** may access the select applications through vehicle's **100** communication interface.

(54) In some embodiments, vehicle **100** may include autonomous or semi-autonomous vehicle-related functionality or technology that may be used with the present embodiments to replace human driver actions may include and/or be related to the following types of functionality: (a) fully autonomous (driverless); (b) limited driver control; (c) vehicle-to-vehicle (V2V) wireless communication; (d) vehicle-to-infrastructure (and/or vice versa) wireless communication; (e) automatic or semi-automatic steering; (f) automatic or semi-automatic acceleration; (g) automatic or semi-automatic braking; (h) automatic or semi-automatic blind spot monitoring; (i) automatic or semi-automatic collision warning; (j) adaptive cruise control; (k) automatic or semi-automatic parking/parking assistance; (l) automatic or semi-automatic collision preparation (windows roll up, seat adjusts upright, brakes pre-charge, etc.); (m) driver acuity/alertness monitoring; (n) pedestrian detection; (o) autonomous or semi-autonomous backup systems; (p) road mapping systems; (q) software security and anti-hacking measures; (r) theft prevention/automatic return; (s) automatic or semi-automatic driving without occupants; and/or other functionality. In these embodiments, the autonomous or semi-autonomous vehicle-related functionality or technology may be controlled, operated, and/or in communication with vehicle controller **110**.

(55) The wireless communication-based autonomous or semi-autonomous vehicle technology or functionality may include and/or be related to: automatic or semi-automatic steering; automatic or semi-automatic acceleration and/or braking; automatic or semi-automatic blind spot monitoring; automatic or semi-automatic collision warning; adaptive cruise control; and/or automatic or semi-automatic parking assistance. Additionally or alternatively, the autonomous or semi-autonomous technology or functionality may include and/or be related to: driver alertness or responsive monitoring; pedestrian detection; artificial intelligence and/or back-up systems; navigation or GPS-related systems; security and/or anti-hacking measures; and/or theft prevention systems.

(56) Moreover, in addition to biometric data, where vehicle **100** is an autonomous or semi-autonomous vehicle, vehicle controller **110** may interpret sensory information from sensors **105** to determine usage of vehicle **100** by one or more vehicle users (e.g., driver **115**) for each trip undertaken by vehicle **100**. As used herein, “trip” refers to a discrete use of vehicle **100**, from an initial starting point (e.g., starting vehicle **100**) to an end point (e.g., reaching a destination or turning off vehicle **100**). Accurately determining usage of vehicle **100** by driver **115** for a trip may

facilitate collecting and/or generating vehicle-based telematics data associated with driver **115**.
(57) In further embodiments, vehicle controller **110** may interpret the sensory information to identify driver **115** in relation to passengers **120** present in vehicle **100**. For example, vehicle computer device **110** may determine positional information for at least one vehicle user present in vehicle **100** during a trip. Positional information may include a position of a vehicle user, a direction of facing of the vehicle user, a size of the vehicle user, and/or a skeletal positioning of the vehicle user. The position of the vehicle user may include which seat the vehicle user occupies. The direction of facing of the vehicle user may include whether the vehicle user is facing forward, reaching forward, reaching to the side, and/or has his/her head turned. The size of the vehicle user may determine whether the vehicle user is an adult or a child. The size of the vehicle user may also include the vehicle user's height. The skeletal positioning may include positioning of the vehicle user's joints, spine, arms, legs, torso, neck face, head, major bones, hands, and/or feet.
(58) In other embodiments, vehicle controller **110** may use additional and/or alternative vehicle telematics data in addition to biometric data, such as sensor information from sensors **105** within a paired mobile device **125**, to verify which vehicle occupant is the driver **115** when multiple mobile devices **125** pair with vehicle **100** during a single trip. In one example, vehicle controller **110** may use gyroscope and/or accelerometer sensor information from the paired mobile device **125** to identify which side of vehicle **100** each occupant used to enter vehicle **100** and/or exit vehicle **100**. In other words, vehicle controller **110** may access and process data from the gyroscope and/or accelerometer for each mobile device **125** to determine whether the user of the mobile device **125** entered vehicle **100** on the left (e.g., driver) or the right (e.g., passenger). If only two user computer devices **206** are present and vehicle controller **110** determines that a first mobile device **125** is associated with the left side of vehicle **100** and a second mobile device **125** is associated with the right side of vehicle **100**, vehicle controller **110** may record that the user associated with the first mobile device **125** is the driver **115** and the user of the second mobile device **125** is a passenger **120** for the trip. It should be understood that although the left side is associated with a “driver's side” and the right side is associated with a “passenger side” herein, as is the custom in the United States of America, this method is easily applied to other driving customs in which the left side is a passenger side and the right side is the driver's side.
(59) Sensors **105** may detect the current surroundings and location of vehicle **100**. Sensors **105** may include, but are not limited to, radar, LIDAR, Global Positioning System (GPS), video devices, imaging devices, cameras, audio recorders, and computer vision. Sensors **105** may also include sensors that detect conditions of vehicle **100**, such as speed, acceleration, gear, braking, cornering, and other conditions related to the operation of vehicle **100**, for example: at least one of a measurement of at least one of speed, direction rate of acceleration, rate of deceleration, location, position, orientation, and rotation of the vehicle, and a measurement of one or more changes to at least one of speed, direction rate of acceleration, rate of deceleration, location, position, orientation, and rotation of the vehicle. Furthermore, sensors **105** may include impact sensors that detect impacts to vehicle **100**, including force and direction and sensors that detect actions of vehicle **100**, such the deployment of airbags. In the example embodiment, all data (e.g., vehicle telematics data, biometric data, sensor data, mobile device **125** usage data) collected by vehicle controller **110** and mobile device **125** is transmitted to DM computing device **210** in real-time via DM module **130**.
(60) While vehicle **100** may be an automobile in the example embodiment, in other embodiments, vehicle **100** may be, but is not limited to, other types of ground craft, aircraft, and watercraft vehicles.
(61) FIG. **2** depicts a simplified block diagram of an example system **200** for implementing process shown in FIGS. **4**, **8**A, and **8**B. In the example embodiment, system **200** may be used for monitoring driving behavior of a driver in real-time. As described below in more detail, DM computing device **210** (a backend server that is in communication with DM module **130** locally in vehicle **100** as part of a user computer device **206** and/or vehicle controller **110**) may be configured

to (i) detect a user computer device associated with a driver inside a vehicle; (ii) collect state data of the user computer device, the state data including data as to a state of the user computer device during a currently occurring trip, the state being at least one of an inactive state and an active state; (iii) collect vehicle operation data of the vehicle while the driver and the user computer device are inside the vehicle, the vehicle operation data including vehicle telematics data; (iv) compare the collected vehicle operation data and the collected state data of the user computer device to determine whether the user computer device was activated during the currently occurring trip; and/or (v) cause a driving reward to be applied to an account associated with the driver for inactivity of the user computer device during the currently occurring trip, wherein the inactivity is based on the inactive state.

(62) In the example embodiment, user computer devices **206** are computers that include a web browser or a software application, which enables user computer devices **206** to be in communication with DM computing device **210**, vehicle controller **110**, and insurer network **212** using the Internet or other wireless network (e.g., Bluetooth®). More specifically, user computer devices **206** may be communicatively coupled to the Internet through many interfaces including, but not limited to, at least one of a network, such as the Internet, a local area network (LAN), a wide area network (WAN), or an integrated services digital network (ISDN), a dial-up-connection, a digital subscriber line (DSL), a cellular phone connection, and a cable modem. User computer devices **206** may be any device capable of accessing the Internet including, but not limited to, a desktop computer, a laptop computer, a personal digital assistant (PDA), a cellular phone, a smartphone, a tablet, a phablet, wearable electronics, smart watch, or other web-based connectable equipment or mobile devices. In some embodiments, user computer device **206** may also be mobile device **125** and/or vehicle controller **110**, both shown in FIG. **1**.

(63) A database server **202** may be communicatively coupled to a database **204** that stores data. In one embodiment, database **204** may include sensor data such as biometric data of driver **115** and a record as to a state (e.g., inactive and active) of user computer device **206** during a current trip, past trips, and/or trips over a predetermined period of time (e.g., every week, month, or couple of months). In a further embodiment, database **204** may also include payment account information associated with driver **115**, such as a payment account number, a signed direct deposit authorization form, and/or a routing number. Database **204** may further include information, such as accumulated driving rewards associated with driver **115**, reward payout preferences, and/or selected penalty options. In the example embodiment, database **204** may be stored remotely from DM computing device **210**. In some embodiments, database **204** may be decentralized. In the example embodiment, a user, such as driver **115**, may access database **204** via user computer device **206** by logging onto DM computing device **210**, as described herein.

(64) DM computing device **210** may be communicatively coupled with one or more user computer devices **206** via DM module **130**. In some embodiments, DM computing device **210** may also be communicatively coupled with vehicle controller **110** via DM module **130**. In some embodiments, DM computing device **210** may be associated with, or is part of a computer network associated with an insurance provider, or in communication with the insurance provider's computer network, such as insurer network **212**. In other embodiments, DM computing device **210** may be associated with a third party and is merely in communication with the insurance provider's computer network. More specifically, DM computer device **210** is communicatively coupled to the Internet through many interfaces including, but not limited to, at least one of a network, such as the Internet, a local area network (LAN), a wide area network (WAN), or an integrated services digital network (ISDN), a dial-up-connection, a digital subscriber line (DSL), a cellular phone connection, and a cable modem.

(65) In the exemplary embodiment, DM computing device **210** is also in communication with a payment network **214** (e.g., payment processing network) associated with a payment account tied to driver **115**. Payment network **214** may relate to a transaction card system, such as a credit card

payment system using MASTERCARD® interchange network. The MASTERCARD® interchange network is a set of proprietary communications standards promulgated by Mastercard International Incorporated for the exchange of financial transaction data and the settlement of funds between financial institutions that are members of Mastercard International Incorporated. (MASTERCARD® is a registered trademark of Mastercard International Incorporated located in Purchase, New York). Payment processing network **214** includes at least a payment processor for processing payment transactions. Payment processing network **214** may further include an issuer/financial institution associated with driver **115** (where the issuer/financial institution issues payment accounts/cards to a cardholder (e.g., payment account holder) such as driver **115**), an issuer/financial institution associated with DM computing device **210**, and/or an issuer/financial institution associated with an insurance provider, such as insurer network **212**. In embodiments where driver **115** has elected to receive driving rewards, such as a direct deposit to their payment account, DM computing device **210** may initiate payments into a payment account associated with driver **115** by communicating with the driver's **115** issuer/financial institution. In embodiments where driver **115** has elected to participate in a penalty system, DM computing device **210** may be configured to receive automatic payments from a payment account associated with driver's **115** issuer/financial institution. In certain embodiments, an issuer/financial institution associated with insurer network **212** may release payments to a payment account associated with driver **115**. In these embodiments, DM computing device **210** may generate and transmit a report summarizing the accumulated driving rewards to the insurer network **212** for insurer network **212** to use in releasing the driving rewards to driver **115** via direct deposit. In certain embodiments where driver **115** has elected to participate in a penalty system, an issuer/financial institution associated with insurer network **212** may receive automatic payments from a payment account associated with driver **115**. In these embodiments, DM computing device **210** may generate and transmit a report and/or a message to insurer network **212** informing insurer network **212** of DM computing device's **210** detection of driver's **115** user computer device **206** in an active state during an active trip.
(66) DM computing device **210** may be any device capable of accessing the Internet including, but not limited to, a desktop computer, a laptop computer, a personal digital assistant (PDA), a cellular phone, a smartphone, a tablet, a phablet, wearable electronics, smart watch, or other web-based connectable equipment or mobile devices. In still further embodiments, DM computing device **210** may be separate from both vehicle controller **110** and mobile device **125** and merely be in communication with them to monitor driver **115** and vehicle **100**.
(67) In an exemplary embodiment, DM computing device **210** may be used by agency or organization responsible for issuing driving credentials (e.g., driver's licenses or other driving qualifications or endorsements) to determine whether an individual driver qualifies for these driving credentials based on vehicle operation data relating to the individual's driving. DM computing device **210** may analyze vehicle operation data corresponding to a driver to determine one or more values relating to a safety or quality of the driver's driving. For example, the values may include and/or relate to a total amount of time driven, speed (e.g., with respect to a speed limit), acceleration, braking, turning, use or status of the user computing device while driving, use of turn signals, proximity to other vehicles or objects, other driving events, and/or scores derived from and/or calculated based on such data. DM computing device **210** may compare these values to thresholds to determine if the driver qualifies for a driving credential (e.g., a drivers license), and if so, may automatically provide access to a digital document relating to these credentials (e.g., a digital driver's license) within a virtual wallet application associated with the driver (e.g., user computer device **206**). After the virtual document has been issued, DM computing device **210** may continually monitor subsequent vehicle operation data associated with the driver and control access to the virtual document based on whether driving assessment values associated with the driver continue to meet the corresponding thresholds. DM computing device **210** may also use the artificial intelligence model to generate reports and/or recommendations for areas to work on to

assist the driver in qualifying for the driving credentials.

(68) In the exemplary embodiment, a driver desiring to obtain a driving credential may register via DM computing device **210** (e.g., upon obtaining a practice or learner's permit), which may be operated by or on behalf of a Department of Motor Vehicles (DMV) or other entity responsible for issuing driving credentials. During the registration, DM computing device **210** may be configured to store, in database **204**, identities and/or identifiers associated with one or more registered drivers and baseline biometric information for each of the registered drivers. As described above, this baseline biometric information enables DM computing device **210** to determine when a registered driver is operating a vehicle such as vehicle **100**. In addition to biometric data, other data relating to the registered driver may also be stored. This may include demographic data (e.g., age, location) or other data based on which a determination may be made to grant the registered driver any driving credentials. For example, in some embodiments, thresholds used to determine whether a registered driver qualifies may depend on an age of the registered driver (e.g., a sixteen-year-old may need to meet a higher set of standards than an eighteen-year-old).

(69) In the exemplary embodiment, when a driver begins to operate vehicle **100**, DM computing device **210** may receive biometric data from vehicle controller **110** and compare biometric data received from vehicle controller **110** to the baseline biometric information to identify a current driver from the plurality of registered drivers. As described above, vehicle **100** may be configured to collect various types of biometric data, such as image data captured for face identification and/or fingerprint data captured for fingerprint identification. In some embodiments, DM computing device **210** may continuously collect biometric data while vehicle **100** is being operated to verify that the same driver is continuing to operate vehicle **100** and/or determine if a new driver has taken over operation of vehicle **100**. When the current driver is identified, DM computing device **210** may log the trip and any vehicle operation data received during the trip in database **204** in association with the current driver. In some embodiments, in addition to or rather than biometric data, other types of data may be used to identify the current driver, such as a manual entry of data and/or scanning a bar code or other visual code (e.g., on a practice or learner's permit or displayed by user computer device **206**).

(70) In the exemplary embodiment, DM computing device **210** may be configured to collect vehicle operation data of vehicle **100** while the identified driver is in vehicle **100** (e.g., during the trip). The vehicle operation data collected may include various different metrics, such as those described above, which may be used to assess the driver's performance and determine whether the driver has met any qualifications of obtaining a driving credential.

(71) In some embodiments, the collected vehicle operation data may include speed data indicative of a speed of vehicle **100** at a certain time and location. This speed data may be generated by a speedometer, determined based on tracking a geolocation of vehicle **100** (e.g., using GPS), and/or determined in other ways. The speed data may also include a speed limit at the corresponding time and location. For example, DM computing device **210** may perform a lookup to determine a speed limit at a give location at which a speed measurement was made. Thus, at any given time, it may be determined whether vehicle **100** is traveling above or below the speed limit and by how much. This speed data may be used as a factor in determining whether the driver qualifies for a driving credential.

(72) In some embodiments, the collected vehicle operation data may include proximity data obtained from exterior radar sensors, lidar sensors, cameras, and/or other devices capable of determining how close vehicle **100** is to other vehicles and objects. This data may indicate whether the driver is driving safely. For example, the proximity data may indicate whether the driver is maintaining a safe distance between vehicle **100** and a preceding vehicle (e.g., depending on a speed the vehicles are traveling, or may indicate whether vehicle **100** is traveling dangerously close to stationary objects (e.g., roadside objects or debris on the road) at unsafe speeds. Additionally, proximity data may be used to assess parking, for example, by determining an adequate amount of

space is maintained around vehicle **100** during and after parking. This proximity data may be used as a factor in determining whether the driver qualifies for a driving credential.

(73) In some embodiments, the collected vehicle operation data may include interior camera data obtained from an interior camera of vehicle **100**. This interior camera data may indicate whether the driver is engaging in safe or unsafe practices while driving. For example, the interior camera data may indicate whether the driver is maintaining attention to the road, periodically checking mirrors and/or checking mirrors before lane changes, and/or whether the driver is becoming distracted while driving. This interior camera data may be used as a factor in determining whether the driver qualifies for a driving credential.

(74) In some embodiments, the collected vehicle operation data may include turn signal data. For example, vehicle **100** may be configured to report when its turn signals are being operated. This data may be compared with camera data and/or location data to determine whether the driver is activating the turn signals before turning or changing lanes. This turn signal data may be used as a factor in determining whether the driver qualifies for a driving credential.

(75) In some embodiments, the collected vehicle operation data may include other types of data. For example, any of the other types of data described herein, such as a total amount of time or distance the driver has undergone instruction, state data of a user computing device of the driver and/or other telematics data may be used to determine whether the driver is safely operating vehicle **100** and/or the driver qualifies for a driving credential.

(76) In the example embodiment, DM computing device **210** may be configured to generate, by applying an artificial intelligence model to the collected vehicle operation data stored in association with the first driver, at least one driver assessment value. The driver assessment value may correspond to a parameter such as total amount of time or distance driven, or may be another type of score or value derived from the vehicle operation data that indicates, for example, how safe the driver has driven and/or how well the driver has mastered certain aspects of driving. For example, scores may be provided for a total amount of time driven, speed (e.g., with respect to a speed limit), acceleration, braking, turning, use or status of the user computing device while driving, use of turn signals, proximity to other vehicles or objects, other driving events, and/or scores derived from a combination of these factors, such as an overall driving or safety score. The generated driver assessment values may be stored in database **204** in association with the corresponding driver.

(77) DM computing device **210** may be configured to train the artificial intelligence model using sample data (e.g., pairs of sets of sample vehicle operation data and sets of sample driver assessment values that should result from the sample vehicle operation data), historical data, and/or some combination thereof. In some embodiments, the artificial intelligence model may be periodically or continuously retrained as new data (e.g., vehicle operation data and associated outcomes) becomes available. Alternatively, other supervised or unsupervised machine learning techniques may be used to train such a model.

(78) In the exemplary embodiment, DM computing device **210** may be configured to determine a registered driver qualifies for a driving credential by comparing the at least one driver assessment value associated with the driver to at least one threshold corresponding to the driving credential. For example, the thresholds may specify a total time or distance that needs to be driven during driving instruction and/or certain minimum scores relating to driving safety and/or driving abilities (e.g., parking and/or driving in high-traffic areas). In some embodiments, the specific thresholds used to determine whether a particular driver qualify for the driving credential may depend on other information relating to the driver. For example, in some cases, the thresholds may depend on the age of the driver, in which a younger driver (e.g., a sixteen-year-old) may be subject to more stringent thresholds than an older driver (e.g., an eighteen-year-old). In another example, the thresholds may depend on a location where the driver lives and/or has practiced driving. For example, the thresholds may depend on a jurisdiction in which the driver lives, whether the driver lives in an urban, suburban, and/or rural area, and/or whether the driver lives in an area deemed to

be safer or less safe for driving. In some embodiments, the thresholds may be based on a comparison to other drivers. For example, a driver may qualify if their driver assessment values are better than a certain predefined percentage of other drivers. In addition to driving credentials, these driver assessment values and thresholds may be used to determine if the driver qualifies for other driving rewards as described herein.

(79) In the exemplary embodiment, DM computing device **210** may be configured to, in response to a determination that a driver has qualified for a driving credential, provide access to a digital document associated with the driving credential. For example, the driver may have user computer device **206** configured to execute a virtual wallet application. If the driver qualifies for a driving credential, DM computing device **210** may communicate with user computer device **206** to cause a digital document (e.g., a virtual driver license) to automatically be provided within the virtual wallet, thus enabling the driver to obtain instant proof of their driving credential. In some embodiment, the digital document being present in the virtual wallet may enable the driver to unlock or activate (e.g., using an NFC, Bluetooth, or other wireless connection) vehicles to which the driver has authorized access.

(80) In some embodiments, DM computing device **210** may control access to the digital document, such as by locking or removing the digital document from the virtual wallet, based on an ongoing analysis of the driver's driving. For example, DM computing device **210** may continue to collect vehicle operation data after the driver has earned the driving credential to determine updated driver assessment values. If the updated driver assessment values fail to meet the required thresholds, DM computing device **210** may determine to, for example, temporarily suspend or revoke the driving credentials and/or impose certain restrictions (e.g., locations or times of day where the driver can drive unaccompanied) until the thresholds are met. In such cases, DM computing device **210** may automatically remove the digital document from the virtual wallet and/or modify the digital document to indicate that it has a locked or restricted status.

(81) In some embodiments, DM computing device **210** may be further configured to determine a cost, such as a cost associated with obtaining the driving credential, based on the at least one driver assessment value. For example, different fee levels may be associated with obtaining the driving credential and/or the corresponding digital document, each with different associated thresholds, and DM computing device **210** may determine the fee based on which of these thresholds are met by the corresponding driver assessment values. Alternatively, the cost may be calculated as a function of the one or more driver assessment values. Other costs associated with vehicle **100** such as, for example, registration fees, personal property tax, insurance costs, emergency service costs, and/or other costs may be similarly determined or calculated based on the driver assessment values associated with the driver. In cases where these costs are recurring, updated driver assessment values based on new vehicle operation data may be used each time the cost is determined.

(82) In some embodiments, DM computing device **210** may be further configured to generate, by applying the artificial intelligence model to the collected vehicle operation data stored in association with the first driver, at least one recommendation for improving driving skills. For example, DM computing device **210** may determine that, for a driver, a driver assessment value and/or score generated by the artificial intelligence model that is associated with parallel parking needs to be improved before the driver can qualify for driving credentials. DM computing device **210** may then generate a recommendation that the driver practice parallel parking. DM computing device **210** may similarly identify other areas of strength or areas of improvements, such as reducing speeding, leaving greater distance between vehicles, reducing distractions or usage of user computer device **206** during driving, checking mirrors or blind spots, or any other steps to driver safer and potentially qualify for driving credentials. The driver may then focus on these areas when practicing driving. Such recommendations may be displayed within a report provided by the DM to the user, for example, within a web page and/or app accessible through the driver's user computer device **206**. The report may also include driver assessment values, vehicle operation data, charts,

graphs, links to instructional videos, and/or other information that may assist the driver in reaching a driving proficiency necessary for obtaining a desired driver credential. In certain embodiments, the reports may also include data that enables a driver to compare their scores to others, such as average for other practicing drivers in the area or for drivers as a whole. Such recommendations and reports may also be generated after the driver has obtained the desired driver credential, so that the driver may continue with driving skill improvement and maintaining any earned driver credentials. In some embodiments, these recommendations and reports may be generated and updated in real time based on newly available data.
(83) In some embodiments, DM computing device **210** may analyze groups of drivers, such as all the drivers trained by a particular driving instructor in order to assess and/or generate recommendations for the driving instructor. Such an analysis may be performed similarly to the analysis for an individual driver, with the artificial intelligence model instead using vehicle operation data associated with a group of drivers, such as those associated with the particular driving instructor. Based on this data, DM computing device **210** may generate driver instructor assessment values or scores, reports, and recommendations for improvement for the driving instructor. For example, if an instructor's students tend to struggle with parallel parking, DM computing device **210** may generate a recommendation that the driving instructor spend more time with each student practicing parallel parking. In some embodiments, this assessment o a driving instructor may be used, for example, to determine appropriate compensation for driving instructor services and/or made available to the public to assist in selection of a driving instructor.
(84) FIG. **3** depicts an example use case **300** for monitoring a driver **115** of vehicle **100** (both shown in FIG. **1**) during a currently occurring trip (e.g., active trip) using system **200** (shown in FIG. **2**) in accordance with one embodiment of the present disclosure. In particular, example use case **300** illustrates one version of using biometric sensors **105** (e.g., biometric input device) to detect driver **115** and to capture data of driver **115** in real-time.
(85) As described above, vehicle **100** may include autonomous or semi-autonomous technology or functionality that include and/or are related to biometric identification. In the example embodiment, vehicle **100** includes a biometric steering wheel **302**. Biometric steering wheel **302** includes biometric sensors **105** that may be positioned at various locations around the steering wheel of vehicle **100**. Biometric steering wheel **302** is configured to detect the vehicle user as driver **115**. In the example embodiment, a vehicle user places their hands on biometric steering wheel **302**. In some embodiments, biometric sensors **105** may be embedded throughout a rim **312** of biometric steering wheel **302**. In these embodiments, vehicle controller **110** may not only detect a vehicle user as driver **115**, but also continuously monitor whether or not driver's **115** hands are on biometric steering wheel **302** during a trip. For example, vehicle controller **110** may monitor whether one hand or both hands are on biometric steering wheel **302** during an active trip. In this example, driving rewards (e.g., incentives) may be determined based on whether or not driver's **115** hands remain on biometric steering wheel **302** during a trip. Vehicle controller **110** may determine whether driver **115** is distracted by using biometric data received from biometric steering wheel **302**. For example, during an active trip, vehicle controller **110** may detect that device **306** (similar to user computer device **206**, as shown in FIG. **2** and/or mobile device **125**, as shown in FIG. **1**) has switched from an inactive state to an active state. In this example, vehicle controller **110** may confirm that driver **115** is on their device **306** (e.g., smartphone) based on detecting that only one hand is placed on biometric steering wheel **302**. In other embodiments, biometric sensors **105** may be strategically positioned on the spoke **308** and/or central pad **310** of biometric steering wheel **302**. In these embodiments, driver **115** may be prompted to provide biometric data at the start of the trip by pressing finger(s) or hand(s) against the spoke **308** and/or central pad **310**. In other embodiments, driver **115** may be prompted to provide biometric data periodically throughout the currently occurring trip. Additionally or alternatively, depending on the placement of biometric sensors **105**, vehicle controller **110** may prompt driver **115** to place their hand(s) on the dashboard

(not shown) directly in front of driver **115**. In further embodiments, vehicle **100** may be equipped with a biometric input device (e.g., biometric sensors **105**) near the driver **115** separate from the steering wheel. In some embodiments, biometric sensors **105** may include one or more cameras and/or scanners that are configured to take biometric images of driver **115**. For example, biometric sensors **105** may include, but are not limited to, a camera that captures facial images and/or a retinal scanner.

(86) In the example embodiment, vehicle controller **110** may detect that a vehicle user is driver **115** by comparing the biometric data received for the active trip to the baseline biometric data stored in a database, such as database **204** (shown in FIG. **2**). The baseline biometric data may include data for only authorized drivers of vehicle **100**. In certain embodiments, vehicle **100** may have different types of biometric data, such as fingerprints, DNA, facial images, retinal images, and/or voice images. In these embodiments, vehicle controller **110** may conduct a multi-step verification process to identify the vehicle user as driver **115**. For example, vehicle controller **110** may receive fingerprint data and a facial image of a user. In this example, vehicle controller **110** may compare both the fingerprint data and the facial image to the baseline biometric data to verify that that vehicle user is driver **115**.

(87) In the example embodiment vehicle controller **110** is also in communication with device **306** of driver **115**. In the example embodiment, device **306** is inside vehicle **100**. Before starting the trip, device **306** connects with vehicle **100** via any wired or wireless connection (e.g., Bluetooth®) over one or more radio links or wireless communication channels. Vehicle controller **110** may determine that a paired device **306** is associated with driver **115** because it is the only device **306** paired with vehicle **100**. In some embodiments, vehicle controller **110** may utilize stored identification data (e.g., IMEI, MEID, ESN) of device **306** to verify that device **306** is associated with driver **115**. In these embodiments, driver's **115** device **306** may be the only user computer device **206** registered with vehicle **100**. In other embodiments, driver **115** may have previously registered device **306** with an insurance provider of the automobile insurance policy for vehicle **100**. In these embodiments, vehicle controller **110** may be in communication with the insurance provider via an insurer network, such as insurer network **212** to access information associated with driver's **115** device **306**. For example, after detecting, from received biometric data, that a vehicle user is driver **115**, vehicle controller **110** may receive, from insurer network **212**, identification data of device **306**. In this example, vehicle controller **110** may compare the received data to information of a paired device **306** to determine whether device **306** is associated with driver **115**.

(88) In further embodiments, vehicle **100** may be in communication with one or more devices **306** that are each associated with one of users of vehicle **100**. In one embodiment, at the beginning and/or at the end of a trip, the application on device **306** may prompt selection of which vehicle user is/was driver **115** of the trip, and vehicle controller **110** may record the selected driver **115** as driver **115** for the trip, such that telematics data associated with that trip may be attributed to the correct driver **115**. This method may be employed as a validation or verification in addition to receiving biometric data of a vehicle user, as disclosed above. For example, after a trip in which driver **115** is determined using biometric sensors **105**, the application on device **306** may prompt confirmation that driver **115** was, in fact, (i) driver **115** of the trip and (ii) associated with the paired device **306** (e.g., vehicle **100** paired with driver's **115** device **306**). Vehicle controller **110** may receive an indication of a positive or negative response to the prompt, and confirm records for the trip. In still other embodiments, vehicle controller **110** may use additional vehicle telematics data in addition to using the received biometric data, such as sensor information from sensors **105** within a paired device **306**, to determine which vehicle user is the driver **115** when multiple devices **306** pair with vehicle **100** during a single trip.

(89) Vehicle **100** also includes a user interface **304** that enables driver **115** to communicate with vehicle **100**. In some embodiments, vehicle **100** includes an infotainment system, which enables driver **115** to engage with applications and functions such as maps, real-time navigation (e.g.,

GPS), and music through vehicle **100** rather than through device **306**. In further embodiments, user interface **304** may display applications from driver's **115** device **306** after connecting with driver's **115** device **306**. For example, prior to the trip, driver **115** may select applications, such as audiobooks from device **306** to be in communication with vehicle controller **110**. In this example, driver **115** may access the selected applications via user interface **304** rather than directly through device **306** (e.g., by unlocking their phone and accessing each application on the phone).
(90) In the example embodiment, upon connecting with device **306**, vehicle controller **110** may automatically activate (e.g., launch) an application that switches device **306** into an inactive mode. For example, the application may turn on a “do not disturb” or driving mode setting on device **306** without driver **115** having to manually select the desired setting on device **306**. In other embodiments, vehicle controller **110** may detect that device **306** is in child mode. Child mode may be a setting or applications offered by device **306** that restricts usage to parent-approved content. In these embodiments, based on the selected setting (e.g., child mode), vehicle controller **110** may determine that someone other than driver **115** is using device **306** in vehicle **100** while driver **115** is driving during the trip. In some embodiments, vehicle controller **110** may confirm that driver **115** is not using device **306** because vehicle controller **110** is continuously receiving biometric data from driver **115** via biometric steering wheel **302**.
(91) FIG. **4** illustrates a flow chart of an example method **400** of monitoring driving behavior of driver **115** in real-time. Method **400** may be implemented by a computing device, such as driver monitoring (DM) computing device **210** (shown in FIG. **2**). In the example embodiment, DM computing device **210** may be in communication with user computer device **206** (shown in FIG. **2**), vehicle controller **110**, mobile device **125** (both shown in FIG. **1**), and device **306** (shown in FIG. **3**). DM computing device **210** may also be in communication with sensors **105** such as biometric sensors **105** (shown in FIG. **1**) and insurer network **212** (shown in FIG. **2**).
(92) In the example embodiment, method **400** includes detecting **402** a user computer device associated with a driver inside a vehicle. Method **400** also includes collecting **404** state data of the user computer device, the state data including data as to a state of the user computer device during a currently occurring trip, the state being at least one of an inactive state and an active state. The inactive state may include user computer device **206** being in a child mode (e.g., kid friendly parent-approved applications), hands-free mode (e.g., talking on a phone through vehicle **100**'s sound system), and/or locked mode (e.g., user computer device **206** such as a phone requires input such as passcode, motion, and/or biometric data from driver **115** to open). In the example embodiment, DM computing device **210** is configured to track the state of user computer device **206** in real-time throughout the duration of the currently occurring trip. DM computing device **210** may record, for example, the overall duration and distance for which user computer device **206** is in the inactive state relative to the active state for the currently occurring trip. Method **400** further includes collecting **406** vehicle operation data of the vehicle while the driver and the user computer device are inside the vehicle, the vehicle operation data including vehicle telematics data. Method **400** also includes comparing **408**, by DM computing device **210**, the collected vehicle operation data and the collected state data of the user computer device to determine whether the user computer device was activated during the currently occurring trip. Method **400** further includes causing **410** a driving reward to be applied to an account associated with the driver for inactivity of the user computer device during the currently occurring trip, wherein the inactivity is based on the inactive state.
(93) FIG. **5** depicts an example configuration of user computer device **206** shown in FIG. **2**, in accordance with one embodiment of the present disclosure. User computer device **502** may be operated by a user **501**. User computer device **502** may include, but is not limited to, user computer devices **206**, DM computer device **210** (both shown in FIG. **2**), device **306** (shown in FIG. **3**), mobile device **125**, and vehicle controller **110** (both shown in FIG. **1**). User computer device **502** may include a processor **505** for executing instructions. In some embodiments, executable

instructions may be stored in a memory area **510**. Processor **505** may include one or more processing units (e.g., in a multi-core configuration). Memory area **510** may be any device allowing information such as executable instructions and/or transaction data to be stored and retrieved. Memory area **510** may include one or more computer readable media. In the example embodiment, memory area **510** includes DM module **130**, which is remote from a backend server, such as DM computing device **210**. DM module **130** can be a plug-in to vehicle controller **110** (shown in FIG. **1**), a separate component or module, and/or a software application.

(94) User computer device **502** may also include at least one media output component **515** for presenting information to user **501**. Media output component **515** may be any component capable of conveying information to user **501**. In some embodiments, media output component **515** may include an output adapter (not shown) such as a video adapter and/or an audio adapter. An output adapter may be operatively coupled to processor **505** and operatively coupleable to an output device such as a display device (e.g., a cathode ray tube (CRT), liquid crystal display (LCD), light emitting diode (LED) display, or "electronic ink" display) or an audio output device (e.g., a speaker or headphones).

(95) In some embodiments, media output component **515** may be configured to present a graphical user interface (e.g., a web browser and/or a client application) to user **501**. A graphical user interface may include, for example, an online store interface for viewing and/or purchasing items, and/or a wallet application for managing payment information. In some embodiments, user computer device **502** may include an input device **520** for receiving input from user **501**. User **501** may use input device **520** to, without limitation, input a username and password to access their insurance policy and/or to access information about trip(s) and corresponding rewards.

(96) Input device **520** may include, for example, a keyboard, a pointing device, a mouse, a stylus, a touch sensitive panel (e.g., a touch pad or a touch screen), a gyroscope, an accelerometer, a position detector, a biometric input device, and/or an audio input device. A single component such as a touch screen may function as both an output device of media output component **515** and input device **520**.

(97) User computer device **502** may also include a communication interface **525**, communicatively coupled to a remote-computing device such as insurer network **212** (shown in FIG. **2**). Communication interface **525** may include, for example, a wired or wireless network adapter and/or a wireless data transceiver for use with a mobile telecommunications network.

(98) Stored in memory area **510** are, for example, computer readable instructions for providing a user interface to user **501** via media output component **515** and, optionally, receiving and processing input from input device **520**. A user interface may include, among other possibilities, a web browser and/or a client application. Web browsers enable users, such as user **501**, to display and interact with media and other information typically embedded on a web page or a website from a remote-computing device. A client application may allow user **501** to interact with, for example, DM computing device **210** and/or insurer network **212** (both shown in FIG. **2**). For example, instructions may be stored by a cloud service, and the output of the execution of the instructions sent to the media output component **515**.

(99) FIG. **6** depicts an example configuration of server **202** shown in FIG. **2**, in accordance with one embodiment of the present disclosure. Server computer device **601** may include, but is not limited to, insurer network **212**, DM computing device **210**, and database server **202** (all shown in FIG. **2**). In the example embodiment, server **202** is DM computing device **210**. Server computer device **601** may also include a processor **605** for executing instructions. Instructions may be stored in a memory area **610**. Processor **605** may include one or more processing units (e.g., in a multi-core configuration).

(100) Processor **605** may be operatively coupled to a communication interface **615** such that server computer device **601** is capable of communicating with a remote device such as another server computer device **601**, insurer network **212** vehicle controller **110**, mobile device **125** (both shown

in FIG. **1**), device **306** (shown in FIG. **3**) DM computing device **210**, and user computer devices **206** (shown in FIG. **2**), for example, by using wireless communication or data transmission over one or more radio links or digital communication channels. For example, communication interface **615** may receive requests from user computer devices **206** via the Internet, as illustrated in FIG. **2**.
(101) Processor **605** may also be operatively coupled to a storage device **634**. Storage device **634** may be any computer-operated hardware suitable for storing and/or retrieving data, such as, but not limited to, data associated with database **204** (shown in FIG. **2**). In some embodiments, storage device **634** may be integrated in server computer device **601**. For example, server computer device **601** may include one or more hard disk drives as storage device **634**.
(102) In other embodiments, storage device **634** may be external to server computer device **601** and may be accessed by a plurality of server computer devices **601**. For example, storage device **634** may include a storage area network (SAN), a network attached storage (NAS) system, and/or multiple storage units such as hard disks and/or solid state disks in a redundant array of inexpensive disks (RAID) configuration.
(103) In some embodiments, processor **605** may be operatively coupled to storage device **634** via a storage interface **620**. Storage interface **620** may be any component capable of providing processor **605** with access to storage device **634**. Storage interface **620** may include, for example, an Advanced Technology Attachment (ATA) adapter, a Serial ATA (SATA) adapter, a Small Computer System Interface (SCSI) adapter, a RAID controller, a SAN adapter, a network adapter, and/or any component providing processor **605** with access to storage device **634**.
(104) Processor **605** may execute computer-executable instructions for implementing aspects of the disclosure. In some embodiments, the processor **605** may be transformed into a special purpose microprocessor by executing computer-executable instructions or by otherwise being programmed. For example, the processor **605** may be programmed with instructions, such as those illustrated in FIG. **4**.
(105) FIG. **7** depicts a diagram **700** of components of one or more example computing devices **710** that may be used in system **200** shown in FIG. **2**. In some embodiments, computing device **710** may be similar to DM computing device **210** (shown in FIG. **2**). Database **720** may be coupled with several separate components within computing device **710**, which perform specific tasks. In this embodiment, database **720** may include baseline biometric data **722** for authorized drivers, sample biometric data **724** received from vehicle users of vehicle **100**, insurance policy provisions **726** associated with vehicle **100** and/or driver **115**, driving data **728** associated with an active trip and/or past trip(s). In some embodiments, database **720** may include sensor data, reports concerning the state of user computer device **206** during a trip, and calculations as to rewards earned by driver **115** for driving with user computer device **206** in an inactive state. In some embodiments, database **720** is similar to database **204** (shown in FIG. **2**).
(106) Computing device **710** may include the database **720**, as well as data storage devices **730**. Computing device **710** may also include a communication component **740** for collecting **404** state data of user computer device **206** and for collecting **406** vehicle operation data of vehicle **100** while driver **115** and user computer device **206** are inside vehicle **100** (shown in FIG. **4**). Computing device **710** may further include a comparing component **750** for comparing **408** the collected vehicle operation data and the collected state data to determine whether user computer device **206** was activated (shown in FIG. **4**). Computing device **710** may also include a detecting component **760** for detecting **402** user computer device **206** associated with driver **115** inside vehicle **100** (shown in FIG. **4**). Computing device **710** may also include a processing component **770** for causing **410** a driving reward to be applied to an account associated with driver **115** for inactivity of user computer device **206** (shown in FIG. **4**). Processing component **770** may further assist with execution of computer-executable instructions associated with the system.
(107) FIGS. **8**A and **8**B illustrate a flow chart of an example method **800** of monitoring driving behavior of driver **115**. Method **400** may be implemented by a computing device, such as driver

monitoring (DM) computing device **210** (shown in FIG. **2**). In the example embodiment, DM computing device **210** may be in communication with user computer device **206** (shown in FIG. **2**), vehicle controller **110**, mobile device **125** (both shown in FIG. **1**), and device **306** (shown in FIG. **3**). DM computing device **210** may also be in communication with sensors **105** such as biometric sensors **105** (shown in FIG. **1**) and insurer network **212** (shown in FIG. **2**).

(108) In the exemplary embodiment, method **800** may include storing **802**, in a database, baseline biometric information for a plurality of registered drivers. In certain embodiments, storing **802** may be performed by DM computing device **210** (shown in FIG. **2**).

(109) In the exemplary embodiment, method **800** may further include collecting **804** biometric data generated by one or more biometric sensors of the vehicle. In certain embodiments, collecting **804** may be performed by DM computing device **210** (shown in FIG. **2**).

(110) In the exemplary embodiment, method **800** may further include comparing **806** the collected biometric data to the baseline biometric information to identify a first driver of the plurality of registered drivers that is present in a vehicle. In some embodiments, the biometric data is generated by one or more of a face identification camera or a fingerprint reader installed in the vehicle. In certain embodiments, comparing **806** may be performed by DM computing device **210** (shown in FIG. **2**).

(111) In the exemplary embodiment, method **800** may further include collecting **808** vehicle operation data of the vehicle while the first driver is in the vehicle. In some embodiments, the vehicle operation data is generated by one or more of a speedometer, exterior radar, exterior cameras, interior cameras, or turn signals of the vehicle. In certain embodiments, collecting **808** may be performed by DM computing device **210** (shown in FIG. **2**).

(112) In the exemplary embodiment, method **810** may further include storing **808** the collected vehicle operation data in association with the first driver in the database. In certain embodiments, storing **810** may be performed by DM computing device **210** (shown in FIG. **2**).

(113) In the exemplary embodiment, method **800** may further include generating **812**, by applying an artificial intelligence model to the collected vehicle operation data stored in association with the first driver, at least one driver assessment value. In certain embodiments, generating **812** may be performed by DM computing device **210** (shown in FIG. **2**).

(114) In some embodiments, method **800** may further include generating **814**, by applying the artificial intelligence model to the collected vehicle operation data stored in association with the first driver, at least one recommendation for improving driving skills. In some such embodiments, method **800** may further include causing **816** the user device associated with the first driver to display the at least one recommendation. In certain embodiments, generating **814** and causing **816** may be performed by DM computing device **210** (shown in FIG. **2**).

(115) In the exemplary embodiment, method **800** may further include determining **818** the first driver qualifies for a driving credential by comparing the at least one driver assessment value to at least one threshold. In certain embodiments, determining **818** may be performed by DM computing device **210** (shown in FIG. **2**).

(116) In the exemplary embodiment, method **800** may further include, in response to the determination, providing **820** access to a digital document within a virtual wallet application executed by the user device associated with the first driver, the digital document associated with the driving credential. In certain embodiments, providing **820** may be performed by DM computing device **210** (shown in FIG. **2**).

(117) In some embodiments, method **800** may further include determining **822** a cost associated with obtaining the driving credential based on the at least one driver assessment value. In certain embodiments, determining **822** may be performed by DM computing device **210** (shown in FIG. **2**).

(118) In some embodiments, method **800** may further include receiving **824** subsequent vehicle operation data associated with the first driver after providing access to the digital document. In

certain embodiments, receiving **824** may be performed by DM computing device **210** (shown in FIG. **2**).
(119) In such embodiments, method **800** may further include generating **826**, by applying the artificial intelligence model to the subsequent vehicle operation data associated with the first driver, an updated at least one driver assessment value. In certain embodiments, generating **826** may be performed by DM computing device **210** (shown in FIG. **2**).
(120) In such embodiments, method **800** may further include determining **828** the first driver qualifies to maintain the driving credential by comparing the updated at least one driver assessment value to at least one threshold. In certain embodiments, determining **828** may be performed by DM computing device **210** (shown in FIG. **2**).
(121) In such embodiments, method **800** may further include controlling **830** access to the digital document based on the determination. In certain embodiments, controlling **830** may be performed by DM computing device **210** (shown in FIG. **2**).
(122) In the systems and methods described herein, a processor or a processing element may employ artificial intelligence and/or be trained using supervised or unsupervised machine learning, and the machine learning program may employ a neural network, which may be a convolutional neural network, a deep learning neural network, or a combined learning module or program that learns in two or more fields or areas of interest. Machine learning may involve identifying and recognizing patterns in existing data in order to facilitate making predictions for subsequent data. Models may be created based on example inputs in order to make valid and reliable predictions for novel inputs.
(123) Additionally or alternatively, the machine learning programs may be trained by inputting sample data sets or certain data into the programs, such as image data, text data, report data, and/or numerical analysis. The machine learning programs may utilize deep learning algorithms that may be primarily focused on pattern recognition, and may be trained after processing multiple examples. The machine learning programs may include Bayesian program learning (BPL), voice recognition and synthesis, image or object recognition, optical character recognition, and/or natural language processing-either individually or in combination. The machine learning programs may also include natural language processing, semantic analysis, automatic reasoning, and/or machine learning.
(124) In supervised machine learning, a processing element may be provided with example inputs and their associated outputs, and may seek to discover a general rule that maps inputs to outputs, so that when subsequent novel inputs are provided the processing element may based on the discovered rule, accurately predict the correct output. In unsupervised machine learning, the processing element may be required to find its own structure in unlabeled example inputs. In one embodiment, machine learning techniques may be used to extract data about the computer device, the user of the computer device, the computer network hosting the computer device, services executing on the computer device, and/or other data.
(125) Based on these analyses, the processing element may learn how to identify characteristics and patterns that may then be applied to training models, analyzing transaction and authentication data, and detecting and analyzing risk.
(126) In addition, although various elements of the computer system are described herein as including general processing and memory devices, it should be understood that the computer system is a specialized computer configured to perform the steps described herein for facilitating real time communication across different networks to update user computing devices to display user spending limits.
(127) As will be appreciated based on the foregoing specification, the above-discussed embodiments of the disclosure may be implemented using computer programming or engineering techniques including computer software, firmware, hardware or any combination or subset thereof. Any such resulting computer program, having computer-readable and/or computer-executable

instructions, may be embodied or provided within one or more computer-readable media, thereby making a computer program product, i.e., an article of manufacture, according to the discussed embodiments of the disclosure. These computer programs (also known as programs, software, software applications or code) include machine instructions for a programmable processor, and can be implemented in a high-level procedural and/or object-oriented programming language, and/or in assembly/machine language. As used herein, the terms “machine-readable medium,” “computer-readable medium,” and “computer-readable media” refer to any computer program product, apparatus and/or device (e.g., magnetic discs, optical disks, memory, Programmable Logic Devices (PLDs)) used to provide machine instructions and/or data to a programmable processor, including a machine-readable medium that receives machine instructions as a machine-readable signal. The “machine-readable medium,” “computer-readable medium,” and “computer-readable media,” however, do not include transitory signals (i.e., they are “non-transitory”). The term “machine-readable signal” refers to any signal used to provide machine instructions and/or data to a programmable processor.
(128) This written description uses examples to disclose the invention, including the best mode, and also to enable any person skilled in the art to practice the invention, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the invention is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they have structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal language of the claims.

Claims

1. A driver monitoring (DM) computing device for monitoring driving behavior, the DM computing device comprising a processor in communication with a memory and a vehicle, the processor configured to: store, in a database, baseline biometric information for a plurality of registered drivers; collect biometric data generated by one or more biometric sensors of the vehicle; compare the collected biometric data to the baseline biometric information to identify a first driver of the plurality of registered drivers that is present in a vehicle; continually collect vehicle operation data of the vehicle while the first driver is in the vehicle; continually store the collected vehicle operation data in association with the first driver in the database; continually generate, by inputting the collected vehicle operation data stored in association with the first driver into an artificial intelligence model, at least one real-time driver assessment value, the artificial intelligence model trained based on historical vehicle operation data and one or more sample driver assessment values; continually update the at least one real-time driver assessment value by inputting further vehicle operation data into the artificial intelligence model as the further vehicle operation data of the vehicle is stored in association with the first driver; continually compare the updated at least one real-time driver assessment value to determine the first driver currently qualifies for a driving credential; to when the first driver currently qualifies for the first driving credential, enable a digital document to be presented within a virtual wallet application executed by a user device associated with the first driver, the digital document associated with the driving credential; and when the first driver does not currently qualify for the first driving credential, disable the digital document from being presented within the virtual wallet application.
2. The DM computing device of claim 1, wherein the processor is further configured to generate, by applying the artificial intelligence model to the collected vehicle operation data stored in association with the first driver, at least one recommendation for improving driving skills.
3. The DM computing device of claim 2, wherein the processor is further configured to cause the user device associated with the first driver to display the at least one recommendation.
4. The DM computing device of claim 1, wherein the processor is further configured to determine a

cost associated with obtaining the driving credential based on the at least one driver assessment value.
5. The DM computing device of claim 1, wherein the biometric data is generated by one or more of a face identification camera or a fingerprint reader installed in the vehicle.
6. The DM computing device of claim 1, wherein the vehicle operation data is generated by one or more of a speedometer, exterior radar, exterior cameras, interior cameras, or turn signals of the vehicle.
7. A computer-implemented method for monitoring driving behavior, the computer-implemented method performed by a driver monitoring (DM) computing device including a processor in communication with a memory and a vehicle, the computer-implemented method comprising: storing, in a database, baseline biometric information for a plurality of registered drivers; collecting biometric data generated by one or more biometric sensors of the vehicle; comparing the collected biometric data to the baseline biometric information to identify a first driver of the plurality of registered drivers that is present in a vehicle; continually collecting vehicle operation data of the vehicle while the first driver is in the vehicle; continually storing the collected vehicle operation data in association with the first driver in the database; continually generating, inputting the collected vehicle operation data stored in association with the first driver into an artificial intelligence model, at least one real-time driver assessment value, the artificial intelligence model trained based on historical vehicle operation data and one or more sample driver assessment values; continually updating the at least one real-time driver assessment value by inputting further vehicle operation data into the artificial intelligence model as the further vehicle operation data of the vehicle is stored in association with the first driver; continually comparing the updated at least one real-time driver assessment value to determine the first driver currently qualifies for a driving credential; when the first driver currently qualifies for the first driving credential, enabling a digital document to be presented within a virtual wallet application executed by a user device associated with the first driver, the digital document associated with the driving credential; and when the first driver does not currently qualify for the first driving credential, disabling the digital document from being presented within the virtual wallet application.
8. The computer-implemented method of claim 7, further comprising generating, by applying the artificial intelligence model to the collected vehicle operation data stored in association with the first driver, at least one recommendation for improving driving skills.
9. The computer-implemented method of claim 8, further comprising causing the user device associated with the first driver to display the at least one recommendation.
10. The computer-implemented method of claim 7, further comprising determining a cost associated with obtaining the driving credential based on the at least one driver assessment value.
11. The computer-implemented method of claim 7, wherein the biometric data is generated by one or more of a face identification camera or a fingerprint reader installed in the vehicle.
12. The computer-implemented method of claim 7, wherein the vehicle operation data is generated by one or more of a speedometer, exterior radar, exterior cameras, interior cameras, or turn signals of the vehicle.
13. At least one non-transitory computer-readable media having computer-executable instructions embodied thereon, wherein when executed by a driver monitoring (DM) computing device including at least one processor in communication with a memory and a vehicle, the computer-executable instructions cause the at least one processor to: store, in a database, baseline biometric information for a plurality of registered drivers; collect biometric data generated by one or more biometric sensors of the vehicle; compare the collected biometric data to the baseline biometric information to identify a first driver of the plurality of registered drivers that is present in a vehicle; continually collect vehicle operation data of the vehicle while the first driver is in the vehicle; continually store the collected vehicle operation data in association with the first driver in the database; continually generate, by inputting the collected vehicle operation data stored in

association with the first driver into an artificial intelligence model, at least one real-time driver assessment value, the artificial intelligence model trained based on historical vehicle operation data and one or more sample driver assessment values; continually updating the at least one real-time driver assessment value by inputting further vehicle operation data into the artificial intelligence model as the further vehicle operation data of the vehicle is stored in association with the first driver; continually compare the updated at least one real-time driver assessment value to determine the first driver currently qualifies for a driving credential; when the first driver currently qualifies for the first driving credential, enable a digital document to be presented within a virtual wallet application executed by a user device associated with the first driver, the digital document associated with the driving credential; and when the first driver does not currently qualify for the first driving credential, disable the digital document from being presented within the virtual wallet application.

14. The at least one non-transitory computer-readable media of claim 13, wherein the computer-executable instructions further cause the at least one processor to generate, by applying the artificial intelligence model to the collected vehicle operation data stored in association with the first driver, at least one recommendation for improving driving skills.

15. The at least one non-transitory computer-readable media of claim 14, wherein the computer-executable instructions further cause the at least one processor to cause the user device associated with the first driver to display the at least one recommendation.

16. The at least one non-transitory computer-readable media of claim 13, wherein the computer-executable instructions further cause the at least one processor to determine a cost associated with obtaining the driving credential based on the at least one driver assessment value.

17. The at least one non-transitory computer-readable media of claim 13, wherein the biometric data is generated by one or more of a face identification camera or a fingerprint reader installed in the vehicle.

18. The DM computing device of claim 1, wherein the processor is further configured to train the artificial intelligence model using the historical vehicle operation data and the one or more sample driver assessment values.

19. The DM computing device of claim 18, wherein the historical vehicle operation data and the one or more sample driver assessment values are organized as pairs of correlated data, each of the pairs of correlated data including (i) a set of historical vehicle operation data and (ii) a sample driver assessment value.

20. The DM computing device of claim 18, wherein the processor is further configured to update the artificial intelligence model based on the collected vehicle operation data.